

Software Requirements Specification — Agents Module (Phase 1)

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Table of Contents

- 1. Introduction
 - 1.1 Purpose
 - 1.2 Intended Audience
 - 1.3 Document Conventions
 - 1.4 Intended Use
 - 1.5 Project Scope
 - 1.6 Definitions and Acronyms
 - 1.7 References
- 2. Overall Description
 - 2.1 User Needs
 - 2.2 Product Perspective
 - 2.3 Assumptions and Dependencies
 - 2.4 Constraints
- 3. System Features and Requirements
 - 3.1 Functional Requirements
 - 3.2 User Interfaces
 - 3.3 External Interface Requirements
 - 3.4 Non-Functional Requirements
- 4. Other Requirements
 - 4.1 Database Requirements
 - 4.2 Legal, Regulatory, Compliance
 - 4.3 Internationalization and Localization
- 5. Appendices
 - A. Glossary
 - B. Analysis Models
 - C. Use Cases
 - D. Requirements Traceability Matrix

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, interface, data, and compliance requirements for the **Agents module** of the Future platform — Phase 1

release. It is the authoritative requirements baseline against which the system is implemented, verified, and accepted.

The Agents module is the AI capability of the Future platform: a governed, multi-tenant, audit-complete assistant that lets employees ask questions, surface insights, propose actions, and receive scheduled summaries grounded in the platform's canonical data.

This SRS describes **what** the system must do and the constraints under which it must do it. It is intended to be a stand-alone document; readers do not need any other artifact to understand, verify, or sign off on the requirements herein.

1.2 Intended Audience

Reader	What this document gives them
Project sponsor / executives	The system's purpose, scope, and success criteria
Product manager	The full feature set, prioritisation, and acceptance scope
Engineering team	The complete, unambiguous specification to implement against
Quality assurance / test team	The verification surface; every "shall" statement is a test obligation
Compliance, legal, audit	Privacy, security, retention, and audit obligations the system commits to
Operations / SRE	Reliability, performance, and degradation behaviour the system must exhibit
Customer-facing teams	The capabilities and limits to communicate to tenants

Reading order: §1–§2 are written for any reader; §3.1–§3.2 for product, engineering, QA; §3.3–§4 for engineering, compliance, operations; §5 for everyone needing definitions, models, or use-case detail.

1.3 Document Conventions

- **Requirement keywords** follow IETF RFC 2119 / ISO/IEC Directives Part 2:
 - **shall / must** — a binding, verifiable requirement.
 - **should** — a recommendation; deviation requires documented rationale.
 - **may** — an optional capability.
- **Requirement identifiers** are immutable for the life of this baseline. Each requirement has a stable identifier of the form:
 - **FR-NNN** — functional requirement

- **NFR-NNN** — non-functional requirement
 - **UI-NNN** — user-interface requirement
 - **EI-NNN** — external-interface requirement
 - **DB-NNN** — database requirement
 - **LR-NNN** — legal/regulatory/compliance requirement
 - **I18N-NNN** — internationalization/localization requirement
 - **UC-NNN** — use case
- **Tables** are the primary structural device for requirements. Every requirement appears as a row in a table to support tooling, traceability, and review.
 - **Diagrams** use Mermaid (rendered by GitHub, GitLab, and most modern Markdown viewers). Where diagrams are omitted (e.g. wireframes), narrative prose describes the surface in sufficient detail for review and implementation.
 - **Dollar amounts** are USD unless otherwise stated.
 - **Times** are in 24-hour format; durations in seconds, minutes, hours, or days as labelled; dates are ISO-8601.
 - **Tenant** is the unit of isolation — an entire customer organisation. Inside a tenant, **user** is the unit of identity.

1.4 Intended Use

This SRS is used to:

1. Establish the requirements baseline at sign-off (this document, v1.0).
2. Drive design, implementation, and test planning.
3. Anchor change control — every requirement change becomes a documented amendment.
4. Provide the verification surface for Phase 1 acceptance.
5. Support compliance, audit, and procurement reviews with a single normative artifact.

This document does **not** describe how the system is implemented internally. Implementation, architecture, deployment topology, code organisation, and technology choices are owned by separate engineering artifacts and may evolve without amending this SRS, provided every requirement herein remains satisfied.

1.5 Project Scope

In scope (Phase 1)

The Agents module — a governed, tenant-isolated AI assistant — comprising:

1. **Conversational surfaces.** Global chat surface in the web application; inline copilot surfaces within Planner module screens; asynchronous (scheduled) workflows.

2. **Planner read capabilities.** Personal and team task views, plan status synthesis, role-scoped analysis grounded in Planner tasks joined to the People domain.
3. **Planner write capabilities.** Natural-language task creation, single-task mutations (reassign, reschedule, mark done, split, link to plan), and meeting-transcript action-item extraction.
4. **Tenant knowledge base (RAG).** Tenant-curated reference Q&A — handbook, policies, onboarding documents, internal FAQs — with cited sources and tenant-keyed retrieval.
5. **Execution-mode framework.** Per-conversation Default approvals / Bypass approvals modes, with a non-bypassable floor for bulk, cross-target, and destructive operations.
6. **Approval inbox integration.** Drafted writes land in the platform's existing approval inbox; the Agents module does not own a parallel approval queue.
7. **Scheduled and event-triggered runs.** Opt-in digests and nudges (morning task brief, end-of-week status, stale-task alerts, at-risk milestone notices), running under explicit per-user delegation grants and producing read-only summaries or inbox drafts only.
8. **Tenant administration.** Self-service configuration of model selection, cost ceilings, schedule policy, tool visibility, execution-mode policy, memory policy, and knowledge-base ingestion.
9. **Governance.** Caller-scoped permission inheritance, kernel audit emission for every action, deterministic replay by trace identifier, dollar-denominated cost ceilings at every scope, GDPR right-to-erasure compliance.
10. **Reliability framework.** Retry, circuit breaker, model-degradation ladder, quality canary, cancellation, idempotency, and honesty-of-failure user messaging.

Out of scope (Phase 1)

The following are explicitly excluded from Phase 1 and are not addressed by any requirement in this SRS:

Excluded	Disposition
Modules other than Planner (with People as supporting context)	Time, Hiring, Performance, Finance, etc. integrations are governed by future SRS amendments.
External tenant rollout	Phase 1 is a controlled SETA-internal pilot. External-customer onboarding is a later phase.
Autonomous writes from scheduled or event-triggered runs	Async runs are read-only or inbox-draft only at Phase 1 (see FR-027).
Cross-conversation memory recall (e.g. "you said X yesterday")	Permanently constrained at Phase 1; revisited in a later phase under an explicit privacy review.
Multi-region deployment	Single region (ap-southeast-1) at Phase 1.
Multi-AI-provider failover at runtime	Single provider (OpenAI) at Phase 1; multi-provider routing is engineered as a drop-in but not activated.
OCR ingestion of image-PDF documents into the knowledge base	Text-only ingestion at Phase 1; OCR deferred.
LLM-as-a-judge automated quality scoring gating release decisions	Deferred until a hand-labelled corpus shows $\geq 95\%$ agreement with human raters. Deterministic scorers only at Phase 1.
Slack, Microsoft Teams, or other channel-surface integrations	Notifications flow through the platform notification module's existing channels only.
Event-source triggers from outbox_event (domain-event-driven runs)	The runtime contract is pre-declared so the adapter drops in later, but event-triggered firing is not wired at Phase 1.

Phase 1 success criteria

Criterion	Verification
The Agents module is live and accessible to the SETA-internal pilot population	Production deployment record; pilot user roster
Read-only Q&A over Planner + People + tenant knowledge base demonstrates correct, cited answers	Acceptance demonstration on 2026-05-20; user-acceptance test report
Zero cross-tenant data-exposure incidents during the sprint window	Daily synthetic cross-tenant probe results; security incident register
AI-provider spend remains within the documented forecast envelope	Daily cost reports vs. forecast
Audit trail answers "who asked, what ran, under what authority, with which data" in one query	Acceptance test on a sampled set of production turns
Every "shall" requirement in §3 has a passing verification artifact (test, observation, audit)	Requirements traceability report (Appendix D) reviewed at acceptance

1.6 Definitions and Acronyms

A complete glossary appears in **Appendix A**. The terms below are necessary to read §1–§2.

Term	Definition
Tenant	A customer organisation using the platform — the unit of data isolation.
User	An authenticated identity within a tenant (employee, manager, administrator, auditor).
Module	A functional area of the Future platform (Planner, People, Time, Hiring, etc.).
Agent	The conversational and asynchronous AI capability defined by this SRS.
Turn	One user request and the agent's complete response.
Conversation	A user-scoped thread of turns; the unit on which execution mode is configured.
Execution mode	A per-conversation setting — Default approvals or Bypass approvals — controlling whether single-target writes confirm before running.
Non-bypassable floor	The set of write actions (bulk, cross-target, destructive) that always require approval-inbox confirmation regardless of execution mode.
Approval inbox	The platform's existing in-product approval surface, reused by the Agents module for non-bypassable drafts.
Tenant knowledge base	A tenant-scoped library of reference documents that the agent retrieves passages from and cites in answers.
Scheduled run	A turn that fires on a schedule (e.g. a morning task brief) rather than from an interactive conversation.
Delegation grant	A scoped, expiring, audited authorisation that lets a scheduled or asynchronous run act on behalf of a user.
Audit trail	The end-to-end record of who asked, what the agent did, with which data, and when.
Trace identifier	The unique identifier of a single turn, used to reconstruct it deterministically.
RLS	Row-Level Security — a database mechanism that filters rows based on session context (in this system, tenant identifier).
OIDC	OpenID Connect — the identity protocol used for SSO sign-in.
OTLP	OpenTelemetry Protocol — the protocol used to export observability data.
RAG	Retrieval-Augmented Generation — answering a question by retrieving passages from a corpus and grounding the model's response in them.
SSO	Single Sign-On — federated authentication via the tenant's identity provider.

SSE	Server-Sent Events — a one-way HTTP streaming protocol used to deliver tokens from the agent to the browser.
GDPR	EU General Data Protection Regulation.
PII	Personally Identifiable Information.

1.7 References

Ref	Title
R-1	IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
R-2	IETF RFC 2119 — Key words for use in RFCs to Indicate Requirement Levels.
R-3	OpenTelemetry Generative AI Semantic Conventions (current published version).
R-4	OpenID Connect Core 1.0 specification.
R-5	EU Regulation 2016/679 — General Data Protection Regulation (GDPR).
R-6	OWASP Top 10 (current edition) — referenced for the security threat model.
R-7	OWASP LLM Top 10 (current edition) — referenced for the LLM-specific threat model.
R-8	ISO/IEC 27001 — Information security management systems (referenced for the operational security control framework).
R-9	OpenAI API Reference (current published version) — chat completions, embeddings, structured outputs, moderation endpoints.
R-10	Microsoft Entra ID OIDC integration documentation.
R-11	Web Content Accessibility Guidelines (WCAG) 2.1 Level AA.
R-12	Future platform — Master Service Agreement and Tenant Privacy Policy templates.

2. Overall Description

2.1 User Needs

The Agents module addresses six concrete user needs, expressed in user-need form (who · what · why):

#	User	Need	Why it matters
UN-01	Employee (daily user)	Ask the platform questions in plain language and get correct, sourced answers about their work, their team, and their company	Removes the click-path overhead of finding information across ten screens; democratises analysis-grade access
UN-02	Employee	Express task changes in plain language ("reassign this to Anh and push to Friday") and have the agent draft or run them	Reduces task-management friction; the platform meets the user where they already think in language
UN-03	Manager / Team lead	Get team-level analysis (workload balance, blockers, throughput) without manually assembling spreadsheets	Shifts management time from data plumbing to decisions
UN-04	Department / Org leader	Get cross-team and cross-plan synthesis with role-scoped data; receive scheduled digests rather than pulling them	Operates the organisation from the platform, not a parallel reporting layer
UN-05	Tenant administrator	Configure the agent's behaviour, costs, knowledge sources, and policies without engineering involvement	Ownership of configuration is a hard requirement for multi-tenant operation; engineering tickets do not scale
UN-06	Auditor / Compliance reviewer	Reconstruct what the agent did, on whose authority, with which data, on demand, in a single query	Compliance posture; without this the platform cannot operate in regulated contexts

2.2 Product Perspective

The Agents module is **one module of the Future platform**, not a stand-alone product. It depends on, and operates within, the wider platform ecosystem.

Platform context — what the Agents module relies on:

Capability	Owner	What the Agents module needs
Identity, authentication, session	Platform identity & shell	Authenticated user identity is available on every inbound request; sign-in is handled outside the Agents module
Permission system (role grants, delegations)	Platform governance / kernel	A <code>canDo(user, resource, action)</code> evaluation is invoked on every protected platform call
Tenant isolation (database, vector store)	Platform data layer	Every tenant-scoped table enforces row-level security keyed to the request session
Audit log infrastructure	Platform governance / kernel	The Agents module emits structured audit events; the kernel persists, indexes, and queries them
Approval inbox	Platform — domain modules	The Agents module emits drafted writes; the domain owns the approval workflow surface
Notification system (email, in-app)	Platform notification module	The Agents module emits notification events; delivery channels are owned by the notification module
Planner module (read and write)	Platform — Planner module	The Agents module reads tasks and plans, and proposes/executes write operations through Planner's existing public interface
People module (read)	Platform — People module	The Agents module resolves user identity, ownership, and reporting lines through People's read interface

Module context — what other modules expect from the Agents module:

The Agents module exposes only two surfaces to other platform modules: a read-only query surface for agent state (conversations, drafts, schedules, audit-correlated events) and a structured event stream describing agent activity. The Agents module does not call any other module's internal data structures directly.

External systems — what the Agents module relies on outside the platform:

External system	Role in Phase 1
OpenAI	LLM and embedding provider (chat completions, embeddings, moderation)
Microsoft Entra ID	Tenant identity provider for SSO sign-in
Observability backend	Trace and span storage; queried by operations and audit
Email / messaging gateway	Outbound delivery for scheduled digests, ingestion notifications, draft alerts (via the platform notification module)

Architectural shape (one paragraph, requirements-relevant only). Inbound user requests cross a single trust boundary into the Agents module; the module operates entirely as the calling user, never as a service account; every call to platform domains traverses a single gate that records audit, enforces permission, and accounts for cost; outputs stream back to the user; structured drafted writes flow into the platform's existing approval surfaces. The only persistent state owned by the module is conversation, draft, schedule, knowledge-base, and observability data; all canonical business data remains in the originating module.

2.3 Assumptions and Dependencies

Assumptions (load-bearing for the Phase 1 release; if any becomes false, the affected scope is re-scoped under change control):

#	Assumption	Failure consequence
AS-01	Microsoft Entra is the primary SSO IdP for the SETA pilot tenant; magic link is the documented fallback	Identity surface re-design; Phase 1 schedule slip
AS-02	Three engineers are allocated to Phase 1 — one AI engineer plus two full-stack engineers	Schedule slip; demonstration scope contracts; sprint go-live moves
AS-03	Generative-AI tooling continues to mature at its current trajectory	Per-engineer output drops; Phase 1 capacity plan is revised
AS-04	The platform-wide legacy-decommission deadline of 2026-12-31 holds	Subsequent module integration cadence compresses
AS-05	OpenAI's commercial terms, regional availability, and rate-limit allowances remain consistent with current offer	Provider-tier shift; cost forecast revises; model-degradation ladder revises
AS-06	Tenants enrolled in Phase 1 have appointed an administrator with authority to configure ceilings and policies	Self-service admin features go unused; engineering becomes the configuration surface

Dependencies (external commitments the Phase 1 release relies on):

#	Dependency	Owner
DP-01	Platform identity, permission, audit, and delegation kernel are available and stable	Platform governance team
DP-02	Planner module exposes its read and write interfaces in the form documented for cross-module use	Planner team
DP-03	People module exposes its read interface in the form documented for cross-module use	People team
DP-04	Approval inbox surface accepts drafted-write events in the contracted shape	Domain modules / notification module
DP-05	Notification module supports the notification event types defined in §3.3	Notification module team
DP-06	Production AWS region <code>ap-southeast-1</code> is provisioned and operational	Platform infrastructure team
DP-07	OpenAI API key is provisioned, stored in the platform's secrets manager, and rotated per policy	Platform security team
DP-08	Microsoft Entra tenant integration is configured for the SETA pilot tenant	IT operations
DP-09	Observability backend is provisioned and accepts OTLP exports	Platform observability team
DP-10	Tenant administrators are appointed and onboarded prior to pilot launch	Pilot programme manager

2.4 Constraints

Binding constraints for Phase 1; rationale is documented per constraint.

#	Constraint	Rationale
CN-01	Single AWS region — <code>ap-southeast-1</code> (Singapore)	Multi-region deployment is gated to a later phase based on traffic justification
CN-02	Single AI provider — OpenAI	Multi-provider runtime routing is gated to a later phase, after the first major provider-outage incident or sustained degradation event
CN-03	A single hardware platform standard across the production fleet	Cost, energy efficiency, and supply-chain simplicity
CN-04	Database-enforced tenant isolation on every tenant-scoped table	Application-layer isolation alone is insufficient under the documented prompt-injection threat model
CN-05	No autonomous writes from scheduled or event-triggered runs	Scheduled runs may read or draft to the inbox only; promotion to autonomous writes is gated to a later phase
CN-06	No cross-conversation long-term memory recall	Within-conversation continuity covers the user-experience target without the privacy risk of cross-conversation recall
CN-07	Observability backend vendor is not selected at this baseline	The runtime exports through a vendor-neutral standard (OTLP) so the choice is a deployment-time configuration
CN-08	Caller-permission inheritance — service-account elevation is prohibited	The architectural premise of the system: the agent is never more privileged than the human asking it
CN-09	Costs are denominated in dollars at every accounting layer	Cache-aware pricing makes token accounting an unreliable cost proxy; dollar accounting is the only consistent unit
CN-10	Knowledge-base ingestion at Phase 1 is text-only (markdown, plain text, text-extractable PDF)	Image-PDF / OCR ingestion is gated to a later phase; image content and embedded-image PDFs are out of scope
CN-11	All requirements in this SRS shall be implementable within the Phase 1 sprint window (2026-04-23 → 2026-05-29) or are explicitly marked deferred (see §1.5)	Schedule discipline

3. System Features and Requirements

3.1 Functional Requirements

Functional requirements are organised by **feature**. Each feature is a coherent capability area; each requirement within a feature is a single verifiable statement. A complete identifier list and traceability map is in **Appendix D**.

3.1.1 Feature: Conversational surfaces and output

The system shall expose three surfaces through which users interact with the agent and shall produce outputs in declared, structured shapes.

ID	Requirement	Priority
FR-001	The system shall provide a global chat surface in the web application, accessible to every authenticated user.	Must
FR-002	The system shall provide inline copilot surfaces inside Planner module screens (task detail, plan detail) and shall pass screen context to the agent for intent resolution.	Must
FR-003	The system shall support asynchronous workflows that fire from a schedule and produce read-only summaries or inbox drafts.	Must
FR-004	Each agent response shall declare its output shape — short answer, list, table, narrative, or chart — before the first response token is delivered to the client.	Must
FR-005	The system shall stream the agent's response progressively to the client surface; users shall begin seeing content within the time-to-first-token target stated in §3.4.	Must
FR-006	Outputs that present data sourced from the platform (tasks, plans, knowledge-base passages) shall include source citations sufficient to navigate to the underlying record or document section.	Must
FR-007	The system shall support concurrent open conversations per user; switching between conversations shall preserve each conversation's execution mode, history, and pending drafts.	Must

3.1.2 Feature: Execution modes and approval routing

The system shall support two per-conversation execution modes, with a non-bypassable floor for high-impact actions.

ID	Requirement	Priority
FR-008	Each conversation shall have an execution mode setting selectable by the user from a chat UI control, with two values: Default approvals and Bypass approvals .	Must
FR-009	In Default approvals mode the system shall preview each write as a structured intent in the conversation and shall not execute the write until the user clicks an explicit confirm control.	Must
FR-010	In Bypass approvals mode the system shall execute single-target writes immediately as part of the response, without separate confirmation.	Must
FR-011	In both execution modes the system shall execute reads immediately, without separate confirmation.	Must
FR-012	Each write tool shall declare a non-bypassable property; tools so declared shall always route through the approval inbox regardless of execution mode.	Must
FR-013	The non-bypassable property shall be set on every tool that performs a bulk batch operation, a cross-target write, or a destructive operation.	Must
FR-014	The execution mode of a conversation shall be resolved at turn start and shall not change mid-turn.	Must
FR-015	Tenant administrators shall be able to disable Bypass mode tenant-wide; while disabled, all conversations in the tenant shall behave as if in Default mode regardless of the user's selection.	Must
FR-016	Tenant administrators shall be able to pin individual tools to "always confirm", overriding the tool's own bypassable declaration.	Must
FR-017	When a tool result contains tenant-authored free-text fields, the system shall route every write proposed afterward in the same turn through the approval inbox, regardless of execution mode.	Must
FR-018	If a user denies an inline preview in Default mode, the system shall record the denial as a structured event and shall not execute the proposed write.	Must

3.1.3 Feature: Governance and trust boundary

The system shall enforce caller-scoped permissions, audit every action, and protect against composition-derived disclosure.

ID	Requirement	Priority
FR-019	Every domain action initiated by the agent shall execute under the caller's identity and permission scope; the system shall not elevate to a service account at any point in the call chain.	Must
FR-020	The system shall produce a structured audit event for every call the agent makes against any platform domain.	Must
FR-021	Each audit event shall record: initiator user identifier, tenant identifier, delegation reference (if any), execution mode, confirmation event reference (if any), tool identifier, prompt-version hash, input arguments, output reference, timestamp.	Must
FR-022	The audit trail shall support reconstruction of "who asked, what ran, under what authority, with which data" for any past turn in a single query.	Must
FR-023	A user shall be able to cancel an in-flight turn at any point during its execution.	Must
FR-024	When a write commits before a cancellation signal arrives, the system shall present the truthful timestamp of the committed write to the user and shall not present a fictional rollback message.	Must
FR-025	Tools that return aggregate or analytical results shall declare a minimum group size; queries returning fewer rows than the declared minimum shall be refused or returned in k-anonymised form.	Must
FR-026	Asynchronous and approved-write actions shall execute under scoped, expiring, audited delegation grants; actions without a valid grant shall be refused.	Must
FR-027	Asynchronous runs shall execute as read-only or as inbox-draft only at Phase 1; autonomous writes from scheduled or event-triggered runs shall be prohibited.	Must

3.1.4 Feature: Conversation memory

The system shall maintain coherent within-conversation context, scoped to the caller's permissions, with administrator controls.

ID	Requirement	Priority
FR-028	Within a single conversation the system shall preserve and use prior turns, prior results, and stated preferences so that a user does not have to re-state context that has already been established.	Must
FR-029	The system shall not recall content from a different conversation in the same user's history.	Must
FR-030	The system shall not recall content across tenant boundaries under any circumstance.	Must
FR-031	The agent's memory layers shall not pre-load any fact that the calling user could not retrieve under their own permission scope.	Must
FR-032	Tenant administrators shall be able to configure the active-conversation memory retention period below the 90-day platform default.	Must
FR-033	Tenant administrators shall be able to configure a per-turn prompt-token cap.	Must
FR-034	Tenant administrators shall be able to disable user-preference memory (e.g. format preferences, default views) tenant-wide.	Must
FR-035	Tenant administrators shall be able to override memory profile per surface (global chat, inline copilot, scheduled run).	Should

3.1.5 Feature: Honesty and degradation

The system shall communicate degradation truthfully and shall never substitute fluent prose for a missing or empty answer.

ID	Requirement	Priority
FR-036	When the LLM provider degrades and the runtime falls back to a smaller or less-capable tier, the system shall display a user-visible notice that the experience has changed.	Must
FR-037	When budget, taint, or provider outage forces the system to refuse a request, the system shall display a plain-language refusal stating the reason category (budget, unavailable, refused).	Must
FR-038	The system shall not produce fluent natural-language output covering an empty or missing answer; the absence of an answer shall be surfaced explicitly.	Must
FR-039	When the knowledge-base retrieval subsystem is unavailable, the system shall answer from non-knowledge-base sources only and shall disclose that the knowledge base could not be consulted.	Must

3.1.6 Feature: Approval lifecycle

Drafted writes shall have a defined lifecycle with TTL, revalidation, and idempotency guarantees.

ID	Requirement	Priority
FR-040	Each draft persisted to the approval inbox shall carry a time-to-live; the default shall be 72 hours and shall be overridable per tool.	Must
FR-041	Drafts whose TTL elapses without confirmation shall be auto-rejected; the initiator shall be notified.	Must
FR-042	On confirmation of a draft, the system shall revalidate the precondition state of the underlying domain entity before executing the write.	Must
FR-043	If revalidation of a confirmed draft fails, the system shall record a structured event, shall not execute the write, and shall notify the initiator with an explicit reason.	Must
FR-044	A retried write shall carry an idempotency key per intended side effect; a retry that completes shall return the original outcome and shall not produce a duplicate write.	Must
FR-045	Drafted writes shall record the permission envelope captured at draft time; if the initiator's permissions narrow between draft and execution, execution shall fail under the standard permission-denied path.	Must

3.1.7 Feature: Audit trail and deterministic replay

Any past turn shall be reconstructible from its trace identifier with the same prompt, tool calls, and outcome.

ID	Requirement	Priority
FR-046	The system shall reconstruct any past turn from its trace identifier into the same prompt, the same sequence of tool calls, and the same outcome.	Must
FR-047	The system shall store every prompt content-addressed by hash in an append-only prompt store; identical prompts shall hash identically.	Must
FR-048	The system shall record per-trace canonicalisation rules so that prompt-assembly inputs (token budgets, summary selection, preference snapshots) replay deterministically.	Must
FR-049	A replay that cannot resolve a referenced prompt-store hash shall raise an error and shall not fall back to a fuzzy or near-match prompt.	Must

3.1.8 Feature: Tenant knowledge base (RAG)

The system shall answer questions from a tenant-curated reference library, with cited sources and tenant-keyed isolation.

ID	Requirement	Priority
FR-050	The system shall answer questions grounded in tenant-curated reference material (handbook, policies, onboarding documents, internal FAQs, process guides).	Must
FR-051	Answers grounded in the knowledge base shall cite the source document and section.	Must
FR-052	The knowledge-base retrieval index shall be tenant-keyed; the system shall not perform cross-tenant search and one tenant's content shall never appear in another tenant's results.	Must
FR-053	Tenant administrators shall be able to upload reference documents in markdown, plain text, and text-extractable PDF through the tenant administration interface.	Must
FR-054	The system shall asynchronously chunk, embed, and index uploaded documents and shall make them available for retrieval upon successful ingestion.	Must
FR-055	Tenant administrators shall be able to browse, edit, deprecate, and re-index uploaded documents.	Must
FR-056	The system shall send a notification to the uploading administrator on ingestion completion (success or failure), including any per-document errors.	Must
FR-057	The system shall enforce per-tenant document quotas with a default of 1,000 documents per tenant, raisable per tenant by platform administrators.	Must
FR-058	The system shall enforce a per-document size cap with a default of 5 MB, raisable per tenant by platform administrators.	Must
FR-059	Documents that fail ingestion shall be surfaced to the administrator in the administration interface with a structured failure reason.	Must

3.1.9 Feature: Planner — read capabilities

The agent shall answer questions about Planner tasks and plans, joined to People for ownership and reporting context.

ID	Requirement	Priority
FR-060	The agent shall answer task-view questions over Planner tasks joined to People — including "my open tasks", "what is due this week", "overdue items I own", "what is my team working on", "who is blocked", "who is overloaded".	Must
FR-061	Read results that reference tasks shall include source-task identifiers sufficient for the user to navigate to the underlying task in Planner.	Must
FR-062	The agent shall produce plan-status synthesis for any plan the caller can access — including narrative status, on-track / at-risk / done breakdown, recent changes, stale items (no update within a configurable threshold), and a stand-up digest summary.	Must
FR-063	The agent shall produce role-scoped analysis appropriate to the caller's authority, including: individual-contributor analysis (own throughput, tasks closed vs. opened); team-lead analysis (team workload balance, completion rate by member, blockers by team); department or org leader analysis (plan progress across overseen teams, cross-team dependency risk, throughput comparison across teams).	Must
FR-064	All analytical and aggregate Planner queries shall respect the minimum-group-size composition defence stated in FR-025.	Must

3.1.10 Feature: Planner — write capabilities

The agent shall support natural-language single-target writes (mode-controlled per FR-008–FR-011) and inbox-only batch and cross-target writes.

ID	Requirement	Priority
FR-065	The agent shall create a Planner task from natural-language input — for example "Follow up with Anh about the Q3 brief by Friday" — resolving owner via People, parent plan from current screen or conversation, due date, and dependencies.	Must
FR-066	Task-creation writes shall be mode-controlled per FR-009 / FR-010.	Must
FR-067	The agent shall execute single-task mutations from natural language — reassign, reschedule, mark done, split, link to parent plan — under the requester's authority and shall be mode-controlled per FR-009 / FR-010.	Must
FR-068	The agent shall extract action items from a meeting transcript as a batch of draft Planner tasks; each item shall carry a suggested owner (resolved via People) and a confidence score.	Must
FR-069	Meeting-extracted batches shall route through the approval inbox regardless of execution mode (FR-013 applies — batch and multi-target).	Must
FR-070	The user shall be able to accept or reject each meeting-extracted item individually before batch execution.	Must

3.1.11 Feature: Scheduled and event-triggered runs

The agent shall support opt-in scheduled digests and nudges, executing under explicit delegation grants.

ID	Requirement	Priority
FR-071	Users shall be able to opt in to scheduled digests including: morning task brief, end-of-week status digest, stale-task nudges, at-risk milestone alerts.	Must
FR-072	Each scheduled run shall execute under an explicit per-user delegation grant; runs without a valid grant shall be refused.	Must
FR-073	Scheduled runs shall produce read-only summaries or inbox drafts only and shall not perform autonomous writes (CN-05).	Must
FR-074	Users shall be able to view, modify, pause, and cancel their own schedules from a self-service surface.	Must
FR-075	Tenant administrators shall be able to set a per-tenant maximum number of active schedules per user.	Should

3.1.12 Feature: Tenant administration

Tenant administrators shall be able to configure agent behaviour without engineering involvement.

ID	Requirement	Priority
FR-076	Tenant administrators shall be able to select the active LLM model tier from the platform's catalogue of approved models.	Must
FR-077	Tenant administrators shall be able to configure cost ceilings at per-turn, per-user-per-day, and per-tenant-per-day scopes, denominated in dollars.	Must
FR-078	Tenant administrators shall be able to configure schedule policy — maximum concurrency, allowed schedule windows, and per-schedule cost caps.	Must
FR-079	Tenant administrators shall be able to configure tool visibility — disable specific tools tenant-wide (e.g. disable meeting-action-item extraction).	Must
FR-080	Tenant administrators shall be able to configure execution-mode policy (per FR-015 / FR-016).	Must
FR-081	Tenant administrators shall be able to configure memory policy (per FR-032 / FR-033 / FR-034 / FR-035).	Must
FR-082	All tenant-administration changes shall be audited as structured events with the configuring administrator's identity, the previous value, the new value, and the timestamp.	Must
FR-083	Tenant-administration changes shall take effect within five minutes of being saved, without engineering intervention or platform redeployment.	Must
FR-084	The platform administration role (operator) shall be distinct from the tenant administration role and shall be able to view but not alter tenant configuration values, except where platform-level policy supersedes tenant policy.	Must

3.1.13 Feature: Cancellation and abort

A unified cancellation path shall serve user, system, and timeout cancellations with a typed reason.

ID	Requirement	Priority
FR-085	The system shall provide a single cancellation path for user-initiated, system-initiated (budget, provider outage, quality canary), and timeout-initiated cancellations, distinguished by a typed reason.	Must
FR-086	The cancellation reason shall be one of a closed enumeration: <code>user</code> , <code>timeout</code> , <code>budget</code> , <code>provider_outage</code> , <code>quality_canary</code> . There shall be no "unknown" cancellation reason.	Must
FR-087	A user-initiated cancellation shall reach the next ceasing point in the agent runtime within the latency target stated in §3.4.	Must
FR-088	If a cancellation arrives after a write has committed, FR-024 (truthful timestamp) shall apply.	Must

3.2 User Interfaces

The system exposes four user-interface surfaces. Each is described in narrative form (no wireframes); detailed visual design is owned by the platform design system.

3.2.1 Global chat surface (web application)

ID	Requirement
UI-001	The global chat surface shall be reachable from a persistent control in the platform's primary navigation, available to every authenticated user.
UI-002	The chat surface shall display the user's conversation history grouped by recency and shall allow opening a new conversation or switching to a prior conversation.
UI-003	The chat surface shall display an execution-mode control (Default approvals / Bypass approvals) that is visible at all times and indicates the current mode for the active conversation.
UI-004	The chat surface shall display a turn-cancel control while a turn is in flight; the control shall disappear when the turn ends.
UI-005	The chat surface shall stream tokens progressively and shall render the declared output shape (table, list, narrative, chart) before tokens begin to arrive, so the layout does not visually reflow as content fills in.
UI-006	The chat surface shall render citations (source-task identifiers, source-document references) as navigable links from within the rendered answer.
UI-007	The chat surface shall render inline preview cards for proposed writes in Default mode; each card shall include the proposed action, the affected entities, and explicit Confirm and Decline controls.
UI-008	The chat surface shall display a non-bypassable indicator on any proposed write that is routed through the approval inbox, and shall surface a link to the inbox item.
UI-009	When the system is operating in a degraded tier (per FR-036), the chat surface shall display a non-blocking banner conveying the plain-language degradation message.
UI-010	When the system refuses a request (per FR-037), the chat surface shall present the refusal in a clearly distinct visual style from a normal answer, with the reason category named.

3.2.2 Inline copilot surface (Planner module)

ID	Requirement
UI-011	The Planner module's task-detail and plan-detail screens shall provide an inline copilot surface (a panel or expandable region) reachable without leaving the screen.
UI-012	The inline copilot shall pass the active screen's primary entity (task identifier or plan identifier) to the agent so that "this task" or "this plan" intents resolve without explicit identifiers.
UI-013	The inline copilot shall display the same execution-mode control as the global chat surface and shall obey FR-014 (mode resolved at turn start).
UI-014	The inline copilot shall not display conversation history from the global chat surface; the inline copilot is conversation-scoped to the screen.

3.2.3 Tenant administration surface

ID	Requirement
UI-015	The tenant administration surface shall be accessible only to users granted the tenant-administrator role.
UI-016	The tenant administration surface shall provide controls for every configurable property required by FR-076 through FR-081.
UI-017	The tenant administration surface shall provide a knowledge-base management view supporting upload (FR-053), browse, edit, deprecate, re-index (FR-055), and ingestion-status review (FR-059).
UI-018	The tenant administration surface shall display current cost-ceiling utilisation for the day (per-user, per-tenant) so the administrator can see the spending posture without leaving the screen.
UI-019	The tenant administration surface shall surface a structured audit-event view filtered to administrative changes (per FR-082).

3.2.4 Approval inbox integration

ID	Requirement
UI-020	Drafted writes from the agent shall appear in the platform's existing approval-inbox surface; the agent shall not introduce a parallel inbox.
UI-021	Each agent-emitted draft in the inbox shall display: the proposed action, the affected entities, the requesting user, the conversation reference, the confidence score (if any), and the TTL expiry.
UI-022	The approval-inbox confirmation control shall trigger the standard execution path (per FR-042 / FR-043).

3.2.5 General usability properties

ID	Requirement
UI-023	All user-interface surfaces shall meet WCAG 2.1 Level AA accessibility requirements (R-11).
UI-024	All user-interface surfaces shall render correctly on the supported desktop browser matrix and on the supported mobile browser matrix.
UI-025	All user-interface text strings shall be externalised for localisation (per §4.3).

3.3 External Interface Requirements

3.3.1 Software interfaces

ID	Interface	Direction
EI-001	The system shall integrate with the OpenAI HTTPS API for chat completions (model <code>gpt-5.4</code> for synthesis and reasoning, <code>gpt-5.4-nano</code> for classification and inline operations) and embeddings (model <code>text-embedding-3-small</code>).	Outbound
EI-002	The system shall use the OpenAI structured-output mechanism for any LLM call whose output is consumed programmatically; structured outputs shall validate against a declared JSON schema before downstream use.	Outbound
EI-003	The system shall integrate with OpenAI's content-moderation endpoint for synthesizer outputs and shall block or annotate output that the moderation endpoint flags above the configured threshold.	Outbound
EI-004	The system shall integrate with Microsoft Entra ID over OpenID Connect (R-4) for tenant single sign-on, accepting identity claims as the basis for authentication context.	Inbound (caller identity)
EI-005	The system shall accept identity claims provided by the platform's session layer; the agent module itself shall not perform IdP discovery or token exchange.	Inbound
EI-006	The system shall export observability data (traces, spans, metrics) over the OpenTelemetry Protocol (OTLP) to a vendor-neutral observability backend.	Outbound
EI-007	Span attributes shall conform to the OpenTelemetry Generative AI semantic conventions (R-3) for LLM-related fields, augmented with the platform-specific attributes required to fulfil §3.4 reliability and observability requirements.	Outbound
EI-008	The system shall emit notification events to the platform notification module for: scheduled-run digests, knowledge-base ingestion status, draft TTL expiry, draft execution failure.	Outbound (internal)
EI-009	The system shall consume read APIs of the Planner module (task and plan retrieval) and the People module (user, ownership, reporting-line resolution) through the platform's documented cross-module read interface.	Outbound (internal)
EI-010	The system shall produce drafted-write events to the Planner module's domain command interface; the agent module shall not write to Planner data structures directly.	Outbound (internal)

EI-011	The system shall integrate with the platform kernel for permission evaluation, audit emission, and delegation grant resolution.	Outbound (internal)
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3.3.2 Hardware interfaces

The Agents module is a cloud-resident service; it has no direct hardware interface requirements. Compute, storage, and networking provisioning are owned by the platform infrastructure layer and are subject to constraint CN-03.

3.3.3 Communications interfaces

ID	Requirement
EI-012	All inbound user traffic to the system shall be over HTTPS (TLS 1.2 or higher).
EI-013	Token-streaming responses to client surfaces shall use Server-Sent Events (SSE) over HTTPS.
EI-014	All outbound calls to OpenAI shall be over HTTPS using the provider's published API endpoints; API credentials shall be retrieved from the platform secrets manager at process start.
EI-015	All outbound calls to the platform notification module, kernel, and other internal modules shall use the platform's internal RPC transport with TLS in transit.
EI-016	All observability exports shall use OTLP over gRPC or HTTP per the deployment's exporter configuration.

3.4 Non-Functional Requirements

3.4.1 Performance

ID	Requirement	Target
NFR-001	Time-to-first-token for an interactive turn shall meet a 95th-percentile latency target.	p95 $\leq$ 2.5 seconds
NFR-002	Total turn wallclock for an interactive turn shall be hard-aborted at the configured ceiling.	30 seconds (interactive); 5 minutes (asynchronous)
NFR-003	The system shall sustain a target concurrent-turn load per tenant.	50 concurrent turns per tenant (initial sizing; tunable per tenant)
NFR-004	The system shall enforce a per-turn dollar-cost ceiling, with a configurable default minimum-remaining budget.	Default minimum remaining: \$0.10
NFR-005	The system shall account for cache-read and cache-write tokens at distinct prices on every cost event.	Cache-aware ledger entries on every cost event
NFR-006	Knowledge-base ingestion end-to-end latency per document shall meet a 95th-percentile target.	p95 $\leq$ 60 seconds for documents $\leq$ 1 MB; $\leq$ 5 minutes for documents up to the 5 MB cap
NFR-007	Knowledge-base retrieval latency for top-K queries ($K \leq 8$) shall meet a 95th-percentile target.	p95 $\leq$ 250 milliseconds
NFR-008	A user-initiated cancellation shall propagate to the next ceasing point in the runtime within a sub-second target.	< 1 second from click to next ceasing point

3.4.2 Security and privacy

ID	Requirement
NFR-009	Every tenant-scoped table shall enforce row-level security keyed to the request session's tenant identifier; cross-tenant rows shall be inaccessible at the database layer regardless of application-level logic.
NFR-010	The agent shall execute all domain calls as the calling user; service-account elevation shall not occur at any point in the call chain (CN-08).
NFR-011	Every tool invocation shall produce a kernel audit event correlated by trace identifier and conversation identifier (FR-020).
NFR-012	Secrets — including LLM provider keys, JWT signing keys, and database credentials — shall be stored only in the platform secrets manager. Secrets shall not appear in environment files, container images, source code, or databases.
NFR-013	OpenAI API credentials shall be rotated at least every 90 days, and immediately on suspected compromise.
NFR-014	The system shall protect against prompt injection through database-level isolation, permission-middleware enforcement, and field-level taint propagation (FR-017); a successful injection attempt shall not exceed the caller's permissions.
NFR-015	The system shall protect against composition-derived disclosure through minimum-group-size declarations on aggregate tools (FR-025) and through observability-driven monitoring of unusual aggregation patterns.
NFR-016	The system shall protect against runaway-cost abuse through layered cost ceilings (per-turn, per-user-day, per-tenant-day, per-delegation) and through query rate limits per user.
NFR-017	All personal data ingested by, stored by, or produced by the system shall be classified, retained, and disposed of per the retention table in §4.1.
NFR-018	A daily synthetic cross-tenant probe shall verify that tenant isolation holds across every tenant-scoped table including the knowledge-base index; a successful cross-tenant read shall page on-call.
NFR-019	The system shall not log raw prompt content to non-tenant-isolated stores; observability prompts are redacted by default and stored under tenant-keyed access controls.

3.4.3 Usability

ID	Requirement
NFR-020	All user-interface surfaces shall meet WCAG 2.1 Level AA conformance (R-11).
NFR-021	All user-interface text shall be available in English at Phase 1; the system shall be architected so additional locales can be added without code changes (per §4.3).
NFR-022	The agent shall communicate state changes — degradation, refusal, partial answer, cancellation outcome — in plain language at no greater than secondary-school reading level (per honesty principle FR-036/FR-037/FR-038).
NFR-023	The chat surface shall provide visible feedback within 1 second of a user action (send, cancel, switch mode, switch conversation).
NFR-024	A user shall be able to identify which conversation they are in, what execution mode is active, and whether a turn is in flight at any moment without scrolling.
NFR-025	Error messages presented to users shall include a structured reason category and shall not present internal stack traces or implementation details.

3.4.4 Reliability

ID	Requirement	Target / Behaviour
NFR-026	The system shall be available during business hours.	99.5% during business hours (single-region MVP)
NFR-027	LLM provider transient errors (HTTP 429, 5xx, timeout) shall be retried with exponential backoff and jitter, with at most one retry (two total attempts), in a single layer (no SDK-stacked retries).	Single retry layer; exponential backoff with jitter; honour Retry-After
NFR-028	A per-tool failure shall trip a per-sub-agent circuit breaker; the tool shall be disabled for the remainder of the turn after the configured failure threshold.	Threshold: 2 failures within a sub-agent
NFR-029	When a model tier degrades, the system shall walk an explicit, ordered fallback ladder; each ladder step shall be observable through distinct trace tags; the user shall see a notice when the ladder shifts their experience.	Ladder shape: provider retry → smaller model → partial answer → tenant-wide tier shift → tier-shift on both tiers → hard refuse → daily-budget refuse → KB-degraded
NFR-030	The system shall execute a synthetic quality probe per model tier on a regular interval; the probe shall flag a tier as degraded when a rolling success-rate threshold is breached.	Interval: hourly; degraded threshold: rolling 30-minute success rate < 90%; hard-refuse threshold: rolling success rate < 50% on both tiers
NFR-031	Within-turn read calls with identical canonicalised arguments shall return cached results without re-invoking the underlying tool.	Canonical-hash keyed within-turn read cache
NFR-032	Write tools that may be retried shall persist an idempotency key per intended side effect; a retry of the same key shall return the original result and shall not produce a duplicate.	Persisted by domain; key shape: per intended side effect
NFR-033	The system shall sample traces at a baseline rate plus 100% on triggers (errors, taint, approvals, ceiling breaches).	Baseline 1% + triggered 100%
NFR-034	The system shall emit a structured cost-event ledger per turn, partitioned by tenant, with cache-aware token breakdown.	Per-event ledger with cache-read / cache-write fields
NFR-035	Cancellation, timeout, budget breach, provider outage, and quality canary degradation shall all	Single abort path; closed reason enumeration

	flow through a single abort path with a typed reason (per FR-085 / FR-086).	
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3.4.5 Maintainability and scalability

ID	Requirement
NFR-036	The system shall be designed to absorb additional domain modules (Time, Hiring, Performance, Finance, etc.) without runtime rewrite; per-module work shall be limited to intent and tool declaration.
NFR-037	The system shall support deployment without downtime through rolling deployment of stateless service instances; in-flight turns shall complete or be cancelled cleanly during deployment.
NFR-038	Tenant administration changes (FR-076 through FR-081) shall take effect within five minutes of save without service redeployment.
NFR-039	The system shall support a model-degradation fallback to a smaller model tier and (post-Phase-1) a multi-provider fallback through the same ordered ladder mechanism, without runtime architecture changes.

4. Other Requirements

4.1 Database Requirements

The following database-level requirements apply across all data owned by the Agents module.

ID	Requirement
DB-001	Every persistent data store owned by the Agents module shall be tenant-isolated; every record shall carry an explicit tenant identifier.
DB-002	Every tenant-scoped table shall enforce row-level security; the request session's tenant identifier shall determine row visibility, independently of application-layer logic.
DB-003	The Agents module shall not declare foreign-key references to tables owned by other modules; cross-module references shall be logical and read through documented module facades.
DB-004	The system shall persist conversation, message, draft, schedule, schedule-run, prompt-store, narrative-store, cost-event, knowledge-base document, knowledge-base chunk, knowledge-base embedding, and audit-correlated metadata.
DB-005	The prompt store and narrative store shall be append-only and content-addressed by hash; identical content shall produce identical keys and shall be stored at most once.
DB-006	Each row in tables holding write-tool results shall carry an idempotency key column, unique per (tenant_id, key) , supporting the retry-safety requirement (FR-044).
DB-007	The knowledge-base index shall support per-tenant vector search with retrieval throughput meeting NFR-007; cross-tenant index entries shall not exist.
DB-008	Each cost event shall record dollar-denominated cost broken down by cache-read tokens, cache-write tokens, prompt tokens, and completion tokens; tenant identifier; user identifier; trace identifier; and timestamp.

Retention and lifecycle

Data class	Default retention	Erasure behaviour on user GDPR request
Conversation content (messages, summaries)	90 days active; archive or hard-delete per tenant configuration	Hard-delete content; retain anonymised audit shell
User preferences	Until user-deleted	Hard-delete on erasure request
Tool result cache	TTL declared per tool (typically minutes to hours)	Eviction on next domain mutation; user-scoped purge on erasure
Prompt store and narrative store	Indefinite (content-hashed, deduplicated)	Not personal data; not subject to user erasure
Audit and cost events	90+ days, configurable per tenant for compliance	Retained under documented legitimate-interest; structural fields only
Trace data	≥30 days, vendor-dependent	User-scoped purge required
Knowledge-base documents, chunks, embeddings	Until administrator deletes or deprecates	Not user-personal data; tenant-owned. Deprecated documents excluded from retrieval; shell retained until admin hard-delete.

ID	Requirement
DB-009	The system shall implement the retention table above; retention windows below the platform default may be configured by tenant administrators per FR-032.
DB-010	A GDPR right-to-erasure request shall execute as a single transactional pipeline: hard-delete user content, hard-delete user preferences, purge user-scoped cache, request user-scoped trace purge, retain anonymised audit shells. Partial completion shall be raised as a compliance incident.
DB-011	Database backups shall be encrypted at rest, retained per tenant configuration, and restorable to a point in time within the documented recovery-point objective.

4.2 Legal, Regulatory, Compliance

ID	Requirement
LR-001	The system shall comply with the EU General Data Protection Regulation (R-5), including lawful basis for processing, data minimisation, purpose limitation, transparency, and the data-subject rights of access, rectification, and erasure.
LR-002	The system shall provide a data-subject access mechanism through which a user (or a tenant administrator on behalf of a user) can obtain a structured export of conversation content, preferences, and audit-correlated metadata.
LR-003	The system shall implement the GDPR right-to-erasure pipeline defined in DB-010.
LR-004	All data processing involving personal data shall be performed under a documented lawful basis (legitimate interest for audit, consent or contract for conversation content, etc.); the lawful basis shall be recorded per data class.
LR-005	The system shall maintain a documented record of processing activities sufficient to satisfy GDPR Article 30 obligations.
LR-006	All personal data shall reside in the platform's documented data residency region (ap-southeast-1) at Phase 1; cross-region data transfers shall not occur except through documented sub-processor agreements (e.g. OpenAI processing for inference).
LR-007	The system shall maintain a documented sub-processor list; tenant administrators shall be informed of sub-processor changes per the platform's tenant master service agreement (R-12).
LR-008	The audit trail (FR-020 / FR-021 / FR-022) shall be retained for at least the period required by the most restrictive applicable regulation in tenant scope.
LR-009	The system shall log administrator actions on tenant configuration with sufficient detail to support compliance review (FR-082).
LR-010	The system shall not present model-generated output that includes personal data the calling user is not permitted to view; this is enforced by FR-019 / NFR-009 / NFR-010.
LR-011	The system shall apply provider content-moderation filters (EI-003) to synthesizer output; output flagged above the configured threshold shall be blocked or annotated.
LR-012	The system shall provide a documented incident-response procedure for data breach scenarios that meets GDPR Article 33 notification timelines.
LR-013	Where the system stores tenant content for the purpose of model training, fine-tuning, or evaluation, it shall do so only with explicit documented tenant consent; default behaviour shall be no training use of tenant content.

LR-014	The system shall align operational security controls with ISO/IEC 27001 (R-8) including access control, change management, vulnerability management, and incident management — recognising that formal certification is a platform-level commitment outside the scope of this SRS.
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4.3 Internationalization and Localization

ID	Requirement
I18N-01	All user-interface text shall be externalised in resource files keyed by locale; no end-user-visible text shall be hard-coded in source.
I18N-02	The system shall support English (en) at Phase 1 as the supported user-interface locale.
I18N-03	The system architecture shall support adding additional locales (e.g. Vietnamese <code>vi</code>) without source-code changes; the addition of a locale shall require translation resources only.
I18N-04	The system shall accept user input in any language the LLM provider supports for the active model; the agent's response language shall match the user's input language unless the user explicitly requests another.
I18N-05	All persistent text fields (conversation content, draft content, knowledge-base documents) shall be stored in Unicode (UTF-8).
I18N-06	Date and time presentation shall respect the user's configured timezone and locale-formatting preferences; storage shall always use UTC ISO-8601.
I18N-07	Currency presentation in cost dashboards shall use the platform's denominated currency (USD per CN-09); locale formatting may vary, the underlying value shall not.

5. Appendices

Appendix A. Glossary

The terms below extend the abbreviated glossary in §1.6.

Business and product terms

Term	Definition
Agent	The conversational and asynchronous AI capability defined by this SRS.
Agent-as-a-Service (AaaS)	Future's commercial positioning — an enterprise platform whose differentiating capability is the agent layer.
Tenant	A customer organisation using the platform — the unit of data isolation. SETA itself is the first tenant; external customers join in later phases.
Module	A functional area of the Future platform — Planner, People, Time, Hiring, Performance, Finance, etc.
Capability iteration	A scoped delivery of agent capabilities. The first iteration is read-only Q&A plus tenant knowledge base; subsequent iterations add single-target writes, then asynchronous writes, then multi-step orchestration.
Execution mode	A per-conversation setting — Default approvals or Bypass approvals — controlling whether single-target writes are previewed before execution. Reads always run immediately in both modes.
Default approvals	The execution mode in which the agent previews each write inline and waits for the user's explicit confirm before executing.
Bypass approvals	The execution mode in which the agent executes single-target writes immediately as part of the response.
Non-bypassable floor	The set of write actions (bulk batches, cross-target, destructive) that always require approval-inbox confirmation regardless of execution mode.
Previewed intent	A structured write the agent shows in-chat in Default mode for the user to confirm; not persisted to the draft store.
Drafted action	A write the agent has composed and persisted to the draft store for inbox-path review. Used for non-bypassable writes.
Approval inbox	The platform's existing in-product approval surface where pending items are reviewed and confirmed.
Scheduled run	A turn that fires on a schedule rather than from a live conversation.
Audit trail	The end-to-end record of who asked, what the agent did, with which data, what the user confirmed, and when.
Tenant knowledge base	A tenant-scoped library of reference documents (handbook, policies, FAQs) imported by tenant administrators. Tenant-keyed by construction.
Citation	A reference rendered in an agent answer that points to the source data record or document section grounding the claim.

Composition attack	The exploitation of multiple permitted aggregate queries to derive information about a single individual; defended by minimum-group-size declarations.
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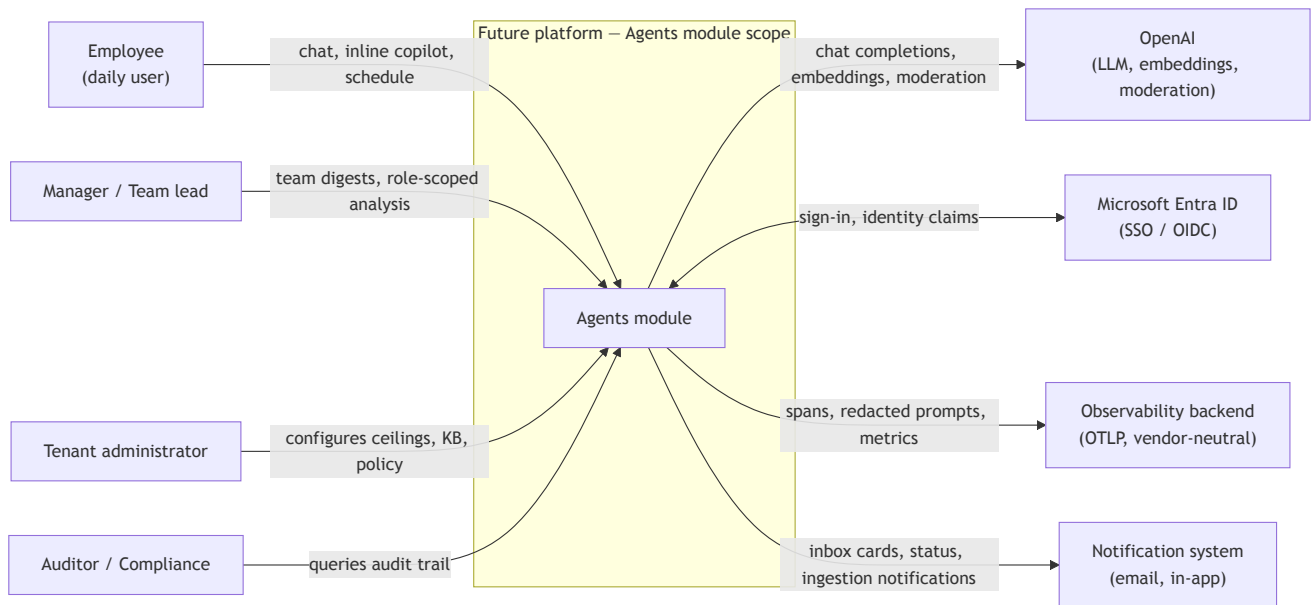
Engineering and runtime terms

Term	Definition
Turn	One user message and the agent's complete response cycle.
Conversation	A user-scoped thread of turns; the unit on which execution mode is configured.
Trace identifier	The unique identifier of a single turn, used to reconstruct the turn deterministically.
Prompt store	An append-only content-addressed store of prompt content, used for replay determinism (FR-046 / FR-047).
Idempotency key	A per-side-effect identifier that ensures a retried write returns the original result rather than producing a duplicate.
Taint flag	A turn-scoped flag set when a tool result returned tenant-authored free-text; once set, every subsequent write in the turn routes through the inbox path regardless of execution mode.
Quality canary	A scheduled synthetic query run against each model tier whose success rate is the runtime's signal of silent quality degradation.
Degradation ladder	An ordered sequence of fallback steps the runtime walks when an LLM provider or model tier degrades; each step is observable and (when the user's experience changes) communicated.
Delegation grant	A scoped, expiring, audited authorisation that lets a scheduled run act on behalf of a user.
Permission envelope	The captured snapshot of a user's permissions at draft-creation time, used by the system to detect permission narrowing between draft and execution (FR-045).

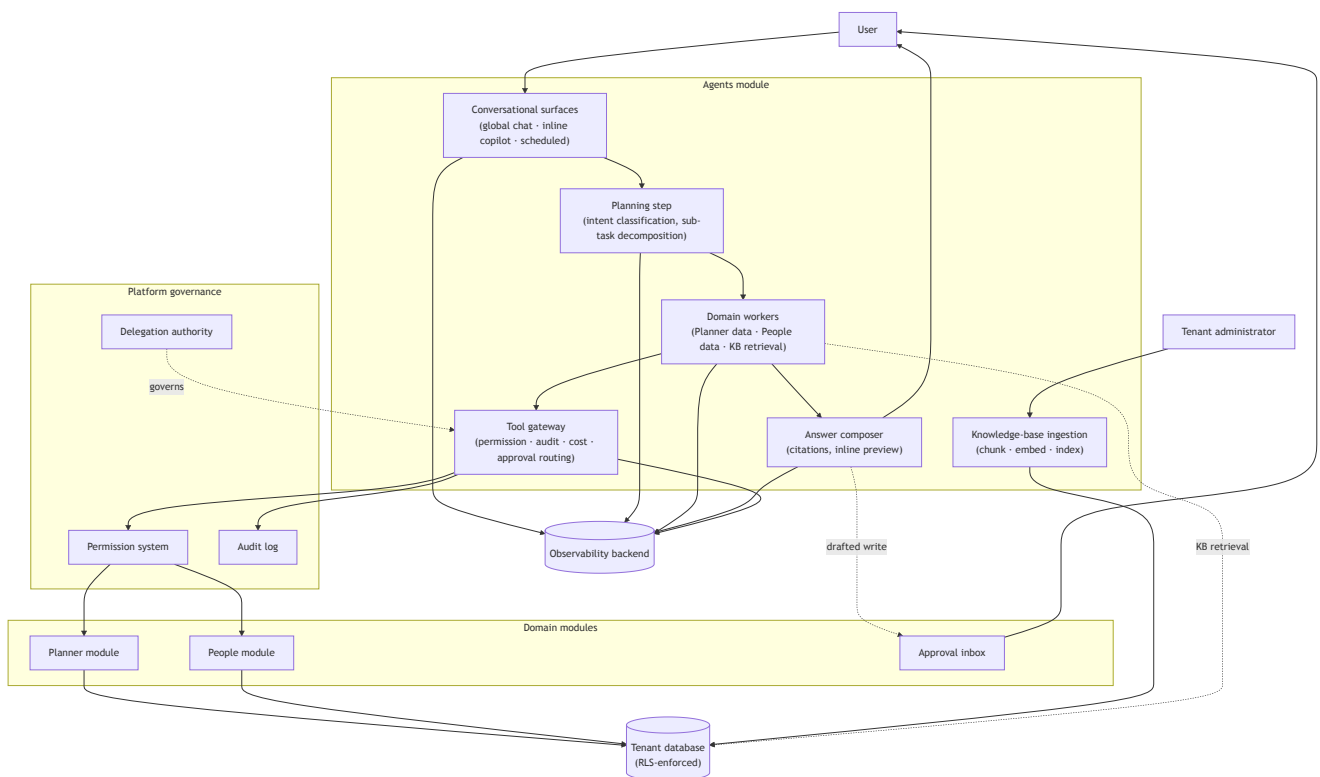
Appendix B. Analysis Models

This appendix presents two analysis models supporting the requirements: the system context view (boundaries and external actors) and the high-level data-flow view (how user requests, agent responses, and audit events flow through the system). These models are illustrative; they do not constitute architectural commitments and may evolve provided the requirements in §3 remain satisfied.

B.1 System context (Mermaid)



B.2 High-level data flow (Mermaid)



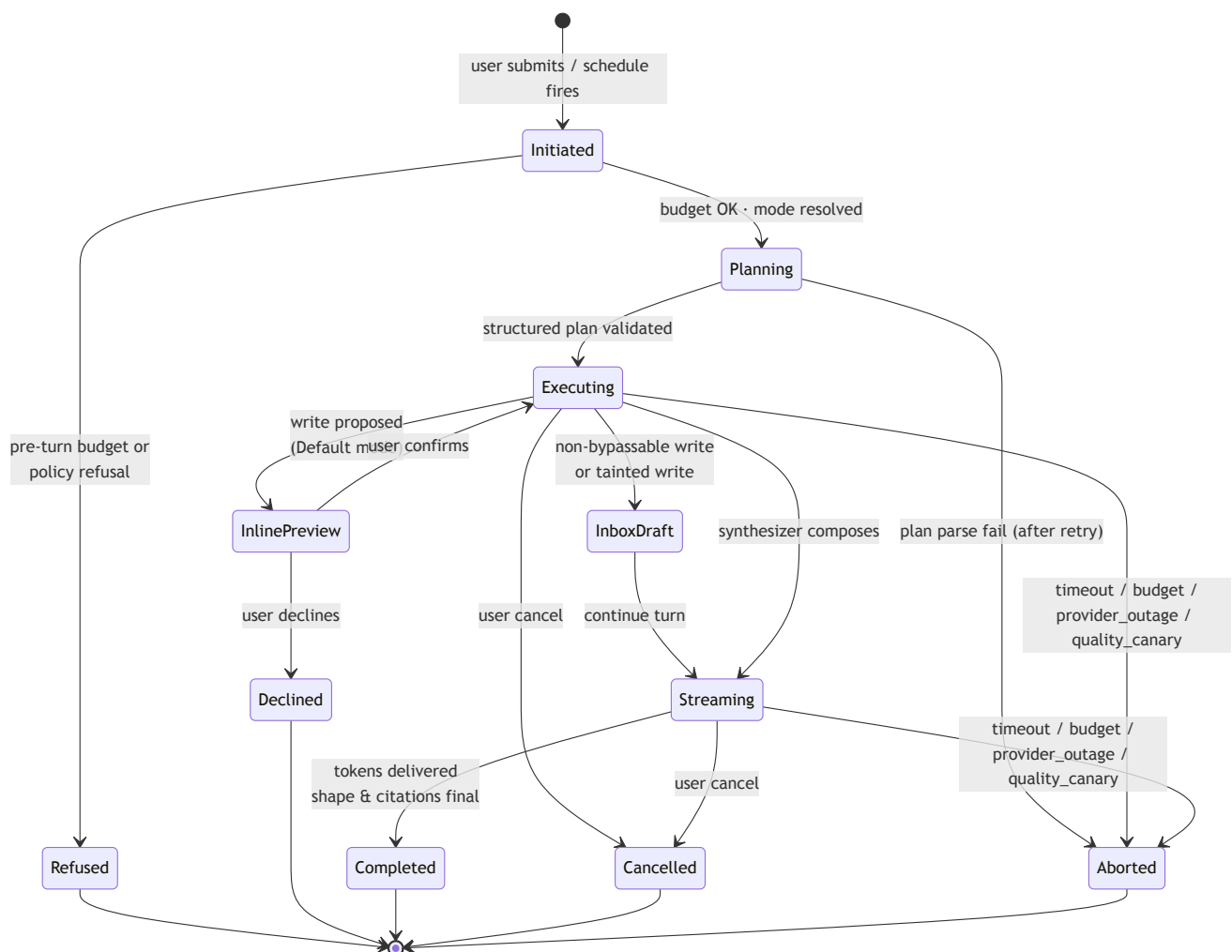
B.3 Trust-boundary properties

The dotted boundary in B.1 is the line every external interaction crosses. Three properties hold at the boundary regardless of which actor or system is on the other side:

1. The caller's identity and permission scope flow into every action the agent takes inside the boundary.
2. Every action that crosses the boundary outbound is audited.
3. No data leaves the boundary except through the four external systems shown.

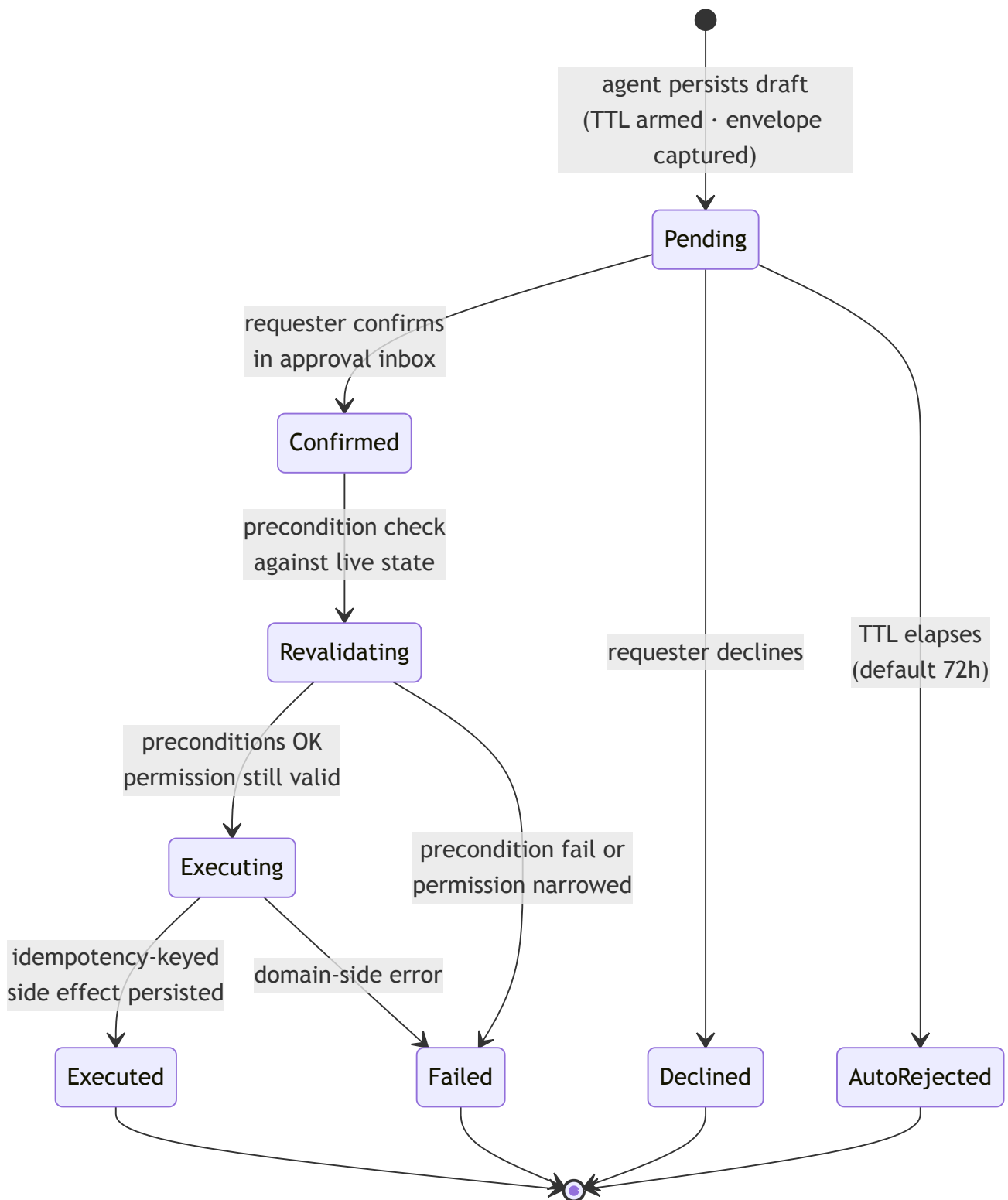
B.4 State diagram — Conversation execution mode and turn lifecycle

A turn is the unit of agent work; each turn passes through a small, deterministic set of states. The lifecycle below is normative for FR-008 through FR-018, FR-023 / FR-024, FR-085 through FR-088, and NFR-026 through NFR-035.



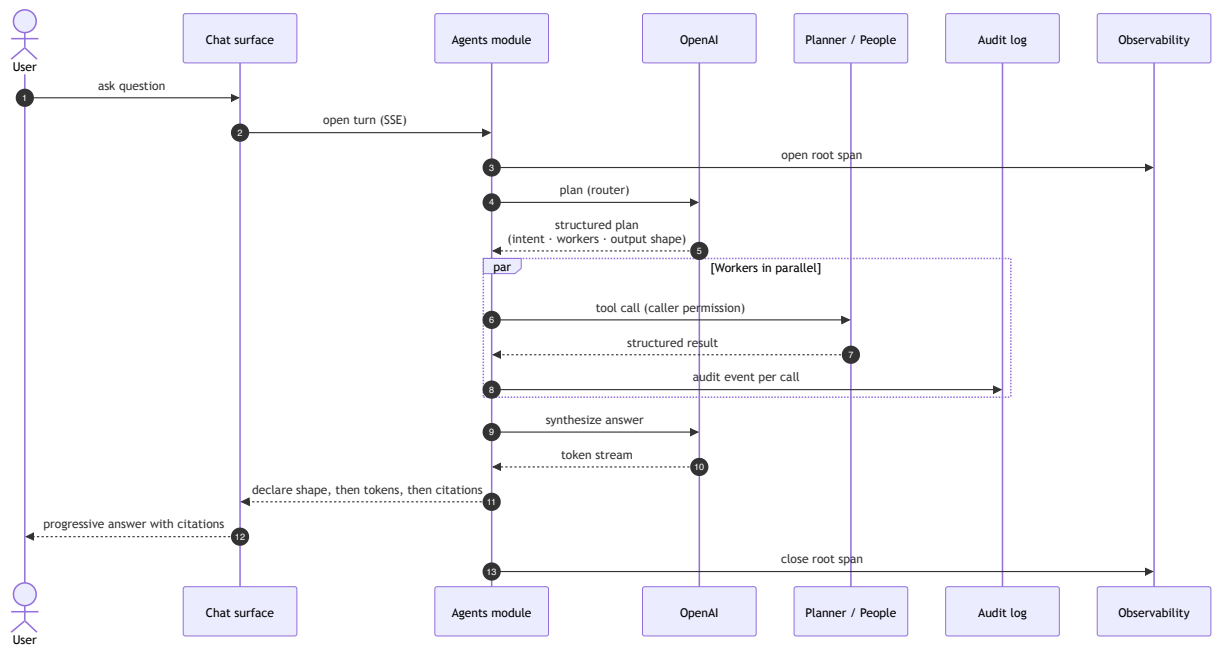
B.5 State diagram — Draft lifecycle

Drafts persisted to the approval inbox follow the lifecycle below. The diagram is normative for FR-040 through FR-045.



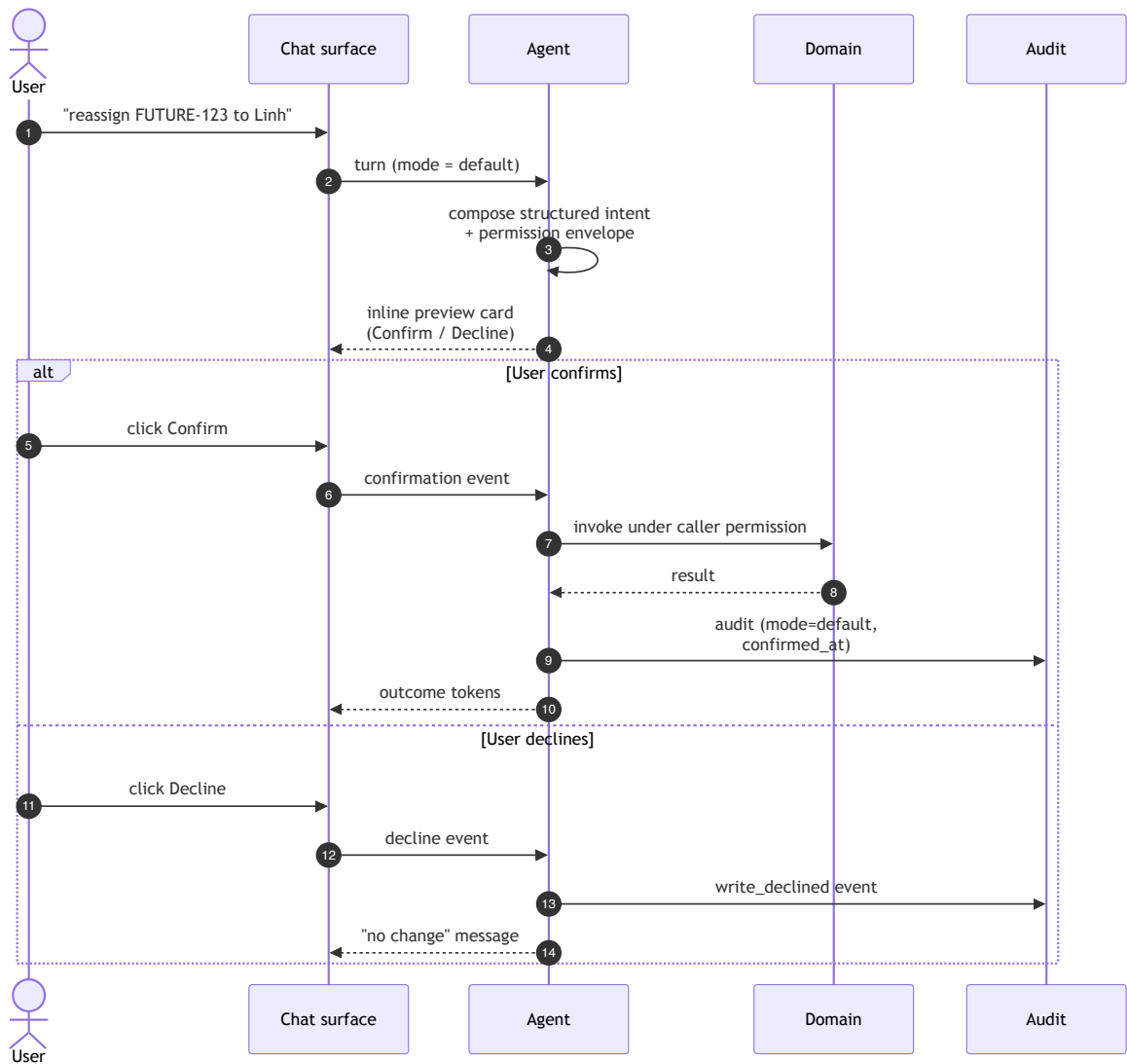
B.6 Sequence diagram — Interactive read-only turn

Canonical flow for FR-001 / FR-005 / FR-019 / FR-020 / FR-060 / FR-062 / FR-063 / NFR-001.



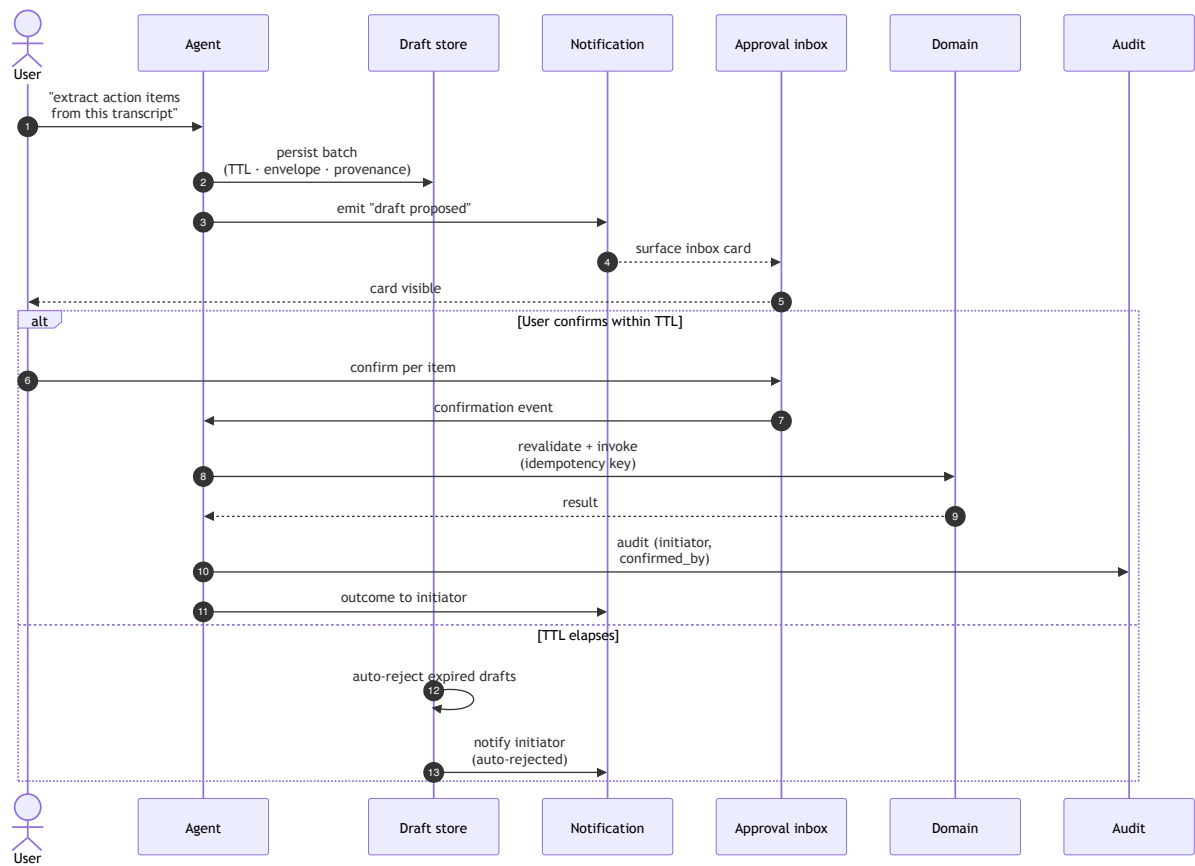
B.7 Sequence diagram — Default-mode single-target write

Canonical flow for FR-008 / FR-009 / FR-018 / FR-021 / FR-067 / UI-007.



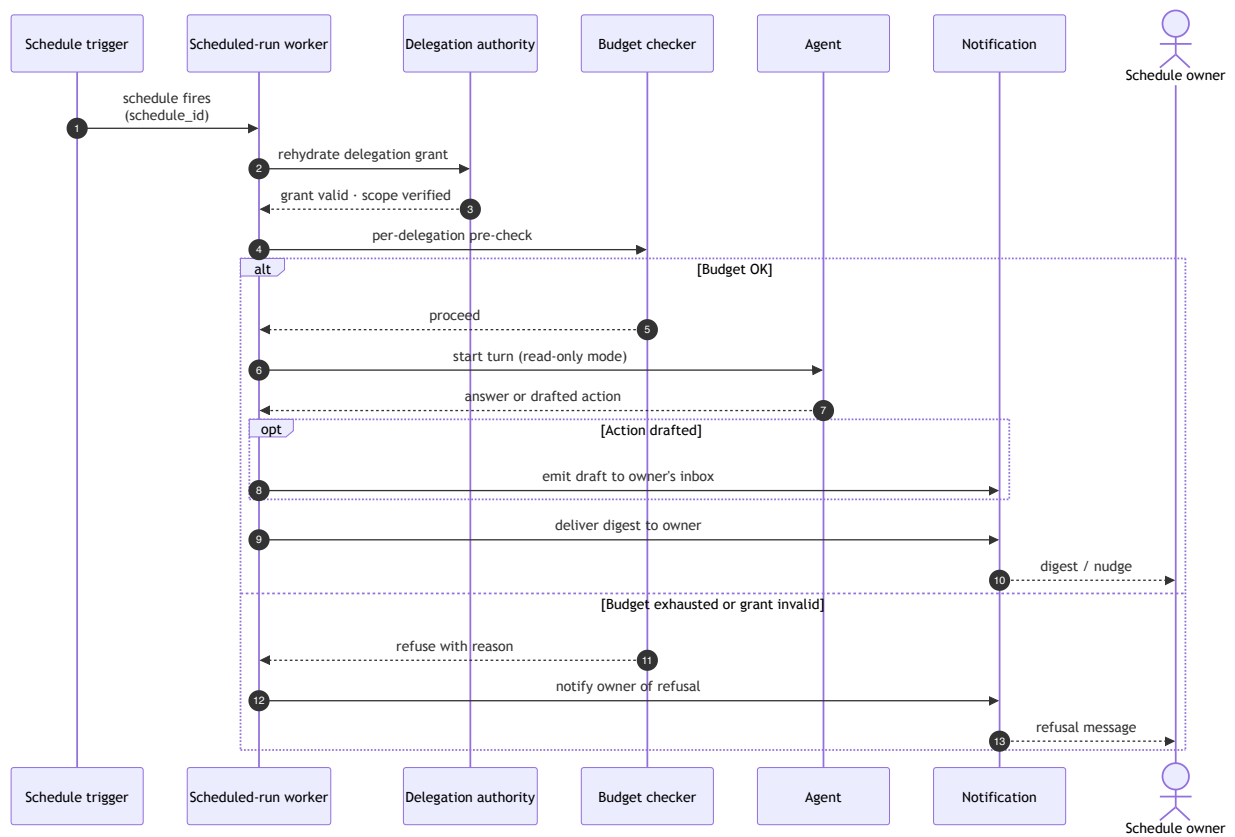
B.8 Sequence diagram — Inbox-path batch write (non-bypassable)

Canonical flow for FR-012 / FR-013 / FR-040 / FR-042 / FR-044 / FR-068 / FR-069 / FR-070.



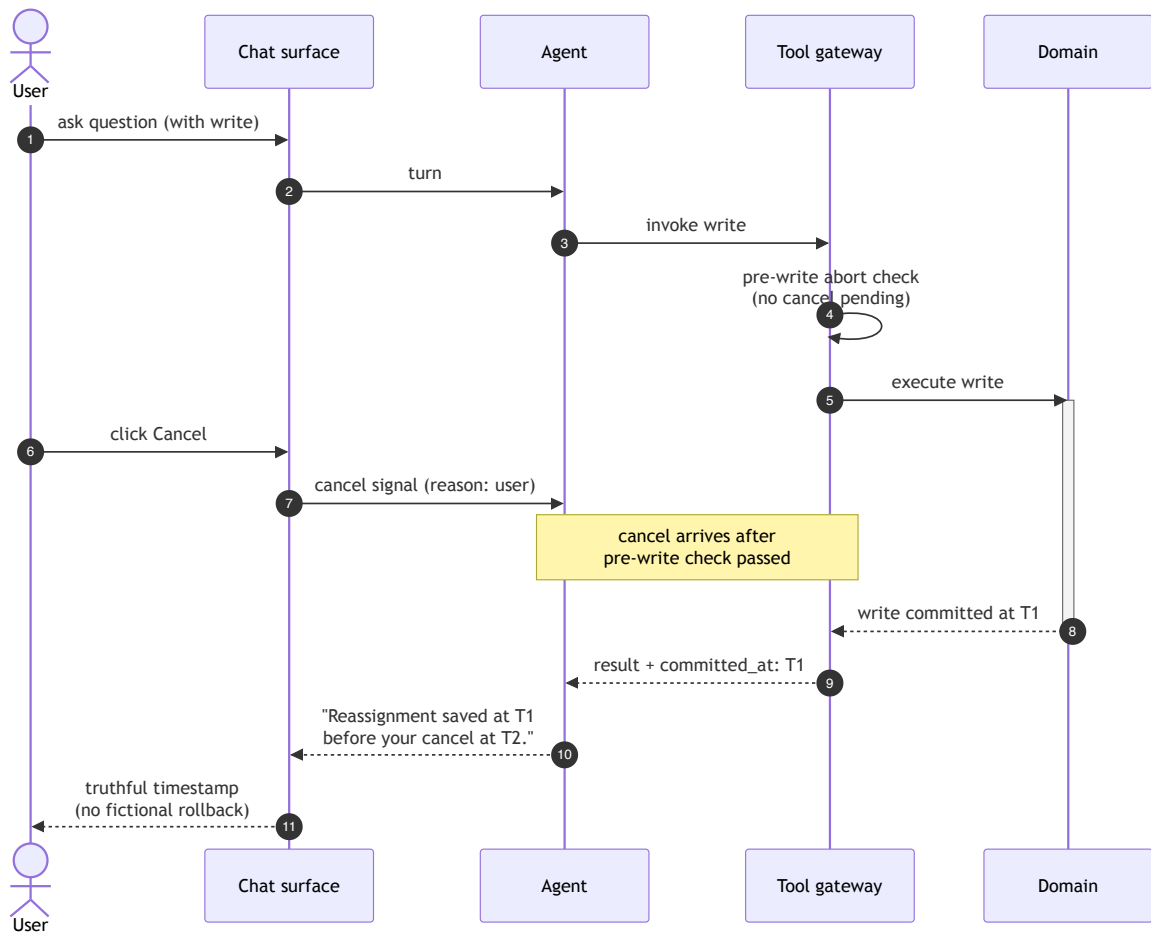
B.9 Sequence diagram — Scheduled asynchronous run

Canonical flow for FR-003 / FR-026 / FR-027 / FR-071 / FR-072 / FR-073.



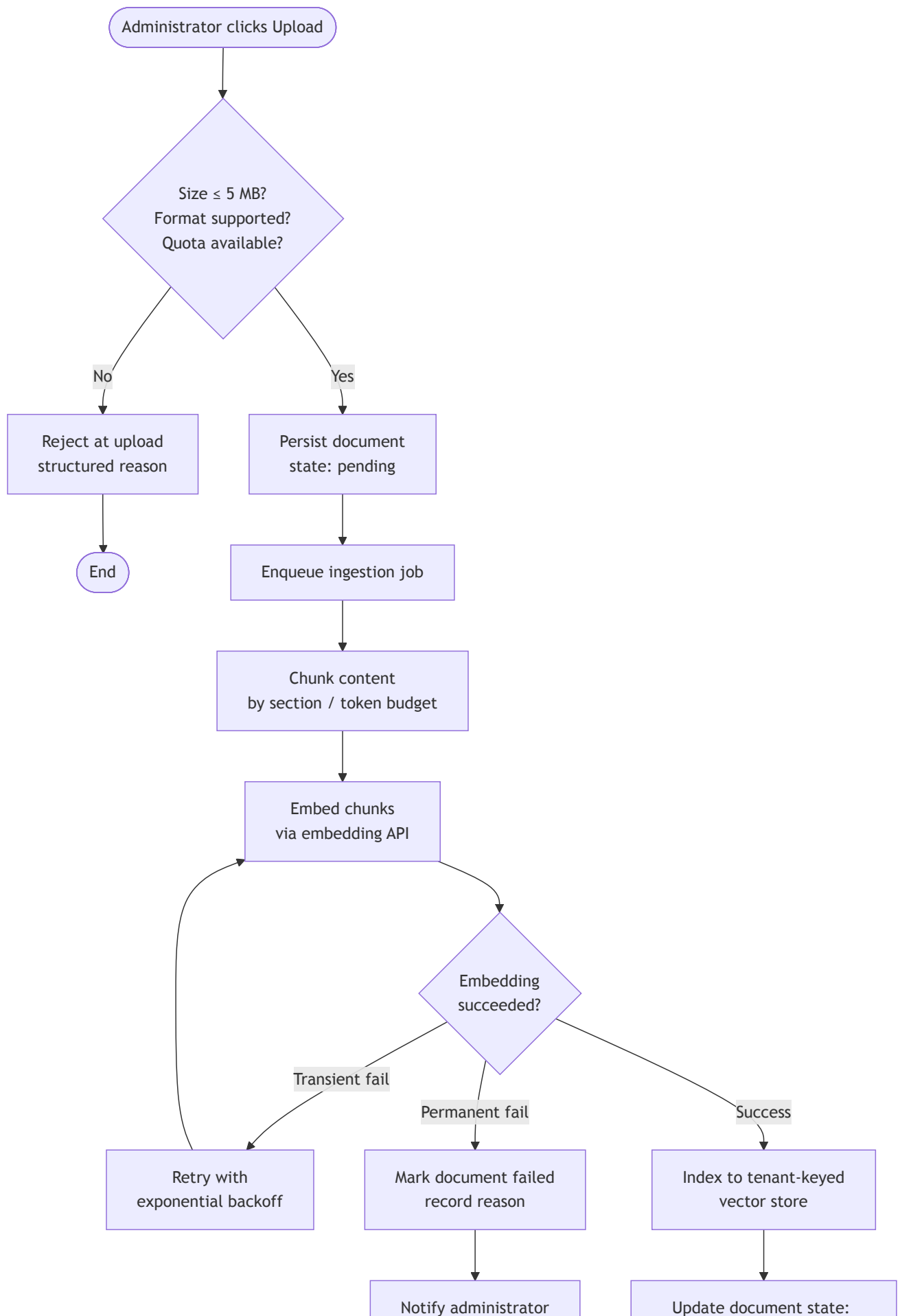
B.10 Sequence diagram — Cancellation with committed-write race

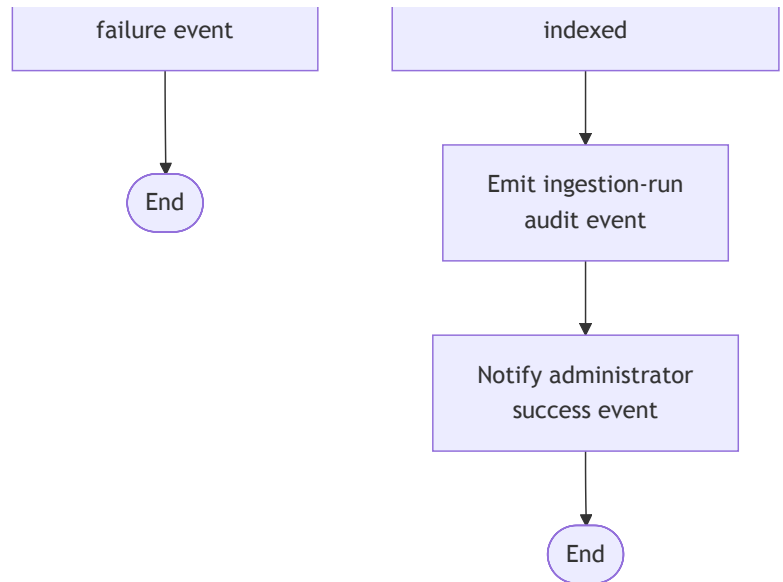
Canonical flow for FR-023 / FR-024 / FR-085 / FR-086 / FR-087 / FR-088 / NFR-008. The diagram shows the honesty contract: when a write commits before the cancel signal arrives, the user is told the truthful timestamp.



B.11 Activity diagram — Knowledge-base ingestion

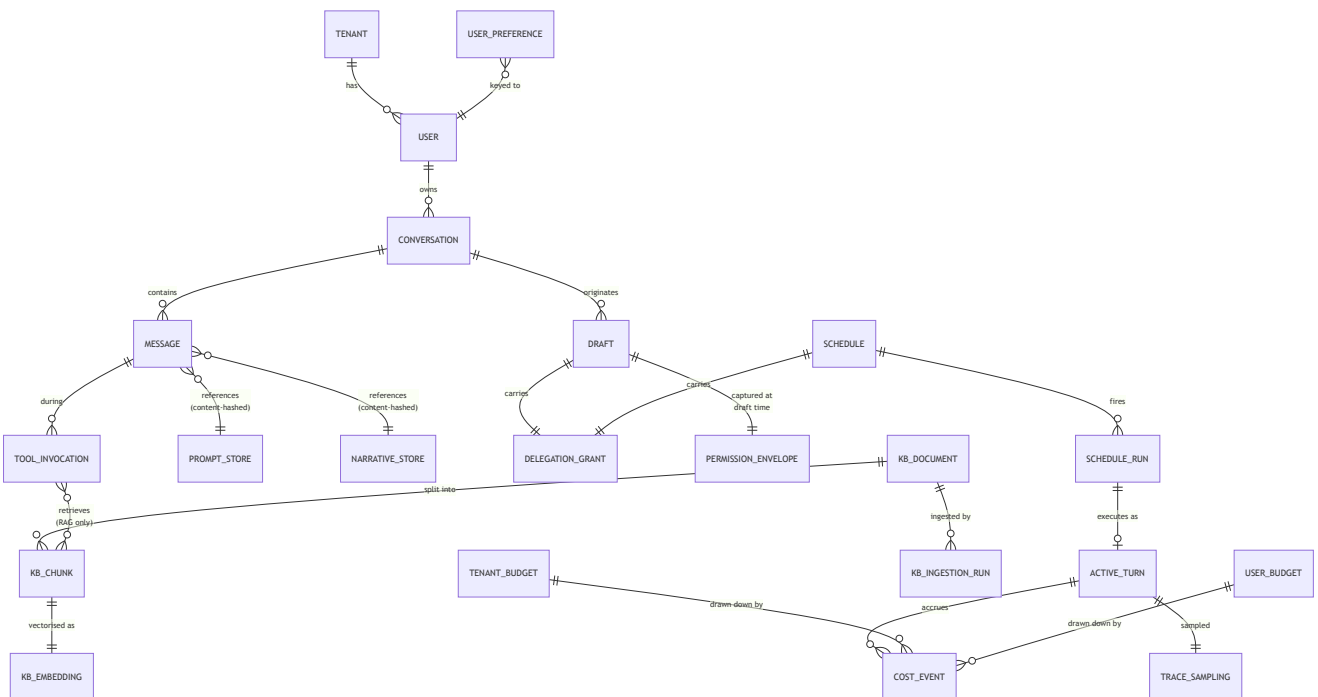
Activity model for FR-053 through FR-059 and NFR-006.





B.12 ER diagram — Logical data model

Logical entity-relationship view of the persistent state owned by the Agents module. The diagram is illustrative for the database requirements in §4.1; physical schema, partitioning, and indexing are owned by separate engineering artifacts.



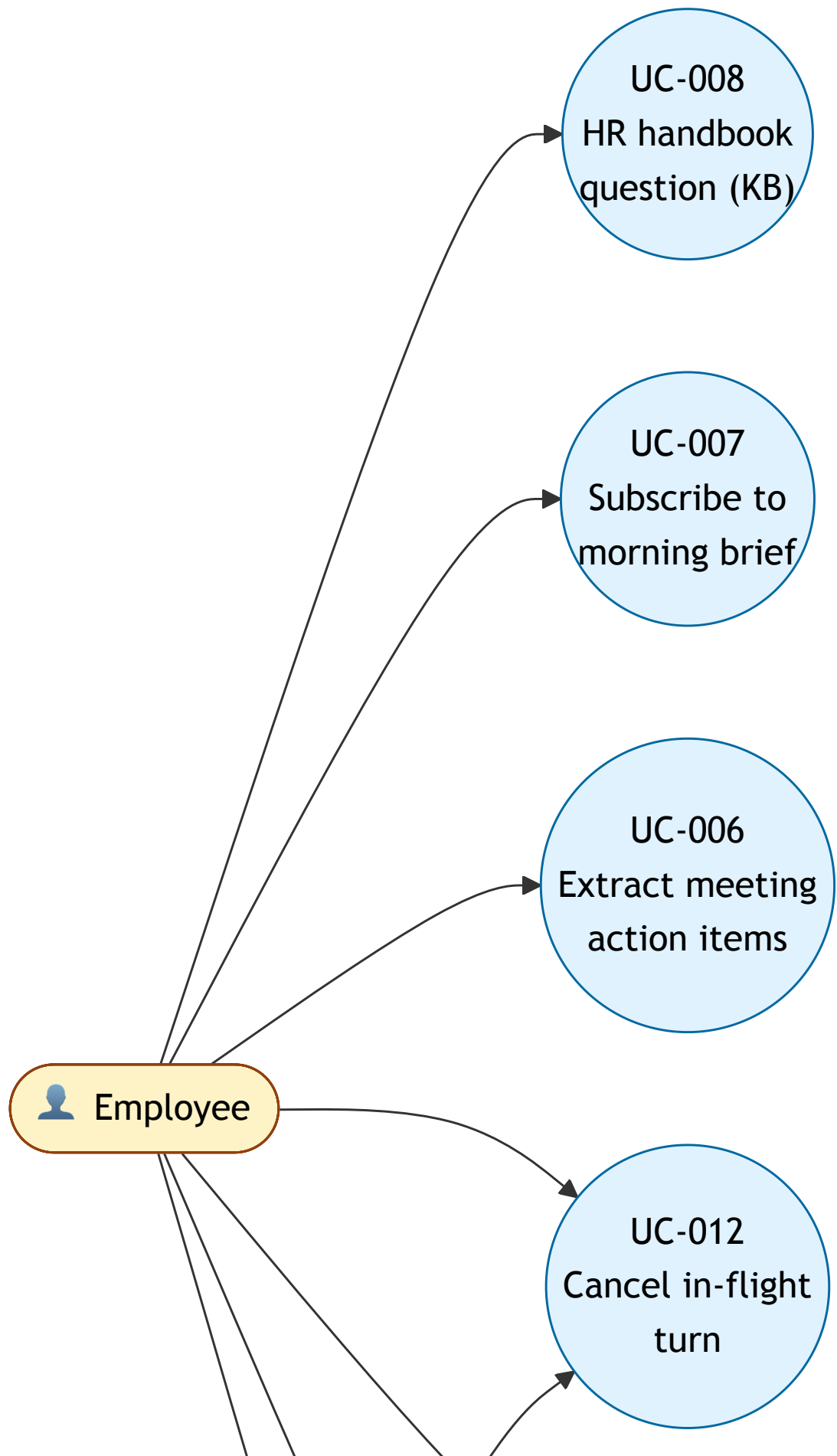
Note on tenant isolation. Every entity above is implicitly keyed by `tenant_id` (per DB-001 / DB-002 / NFR-009). The ER diagram omits the tenant key for visual clarity; in implementation it appears in every entity's primary key.

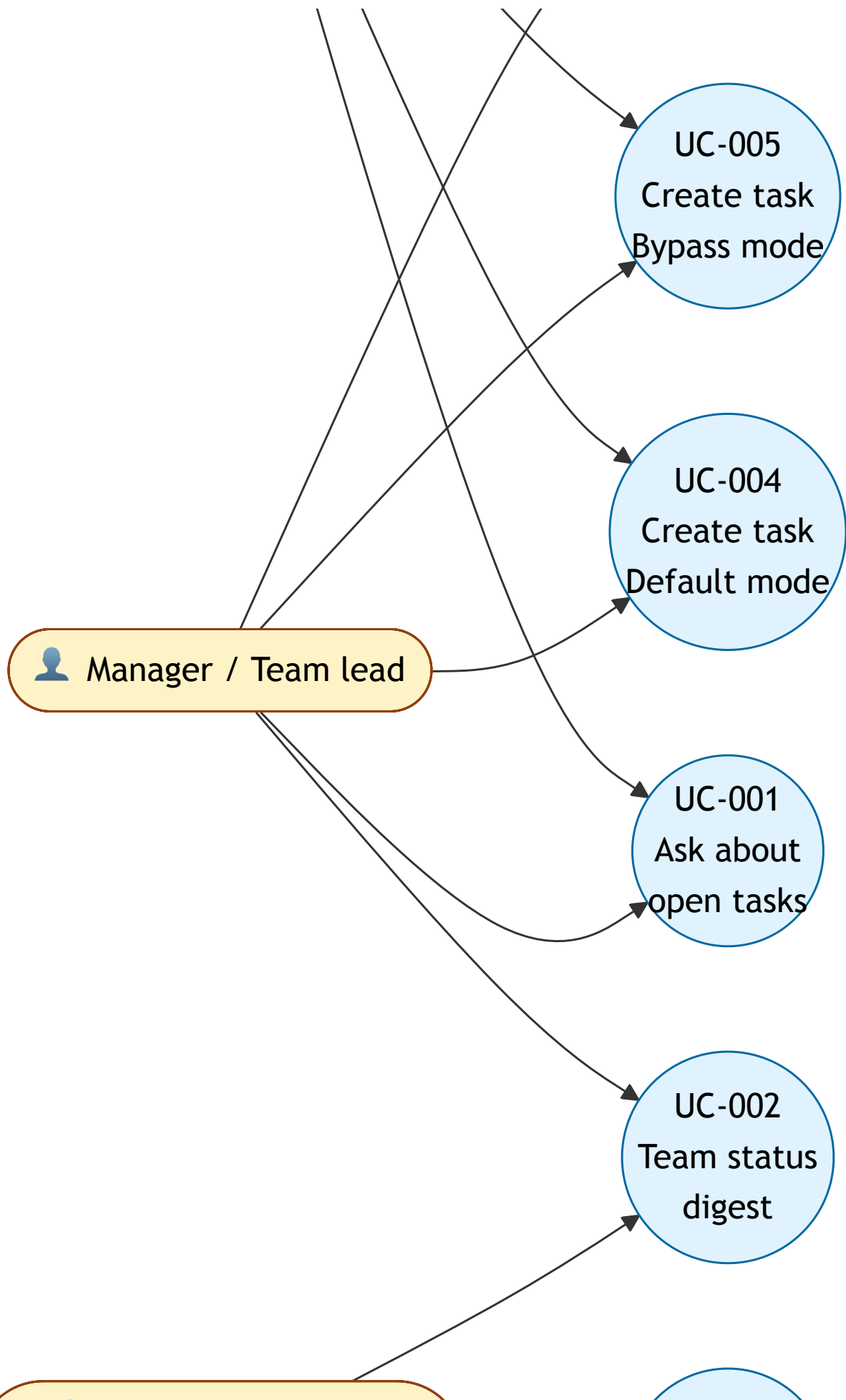
Appendix C. Use Cases

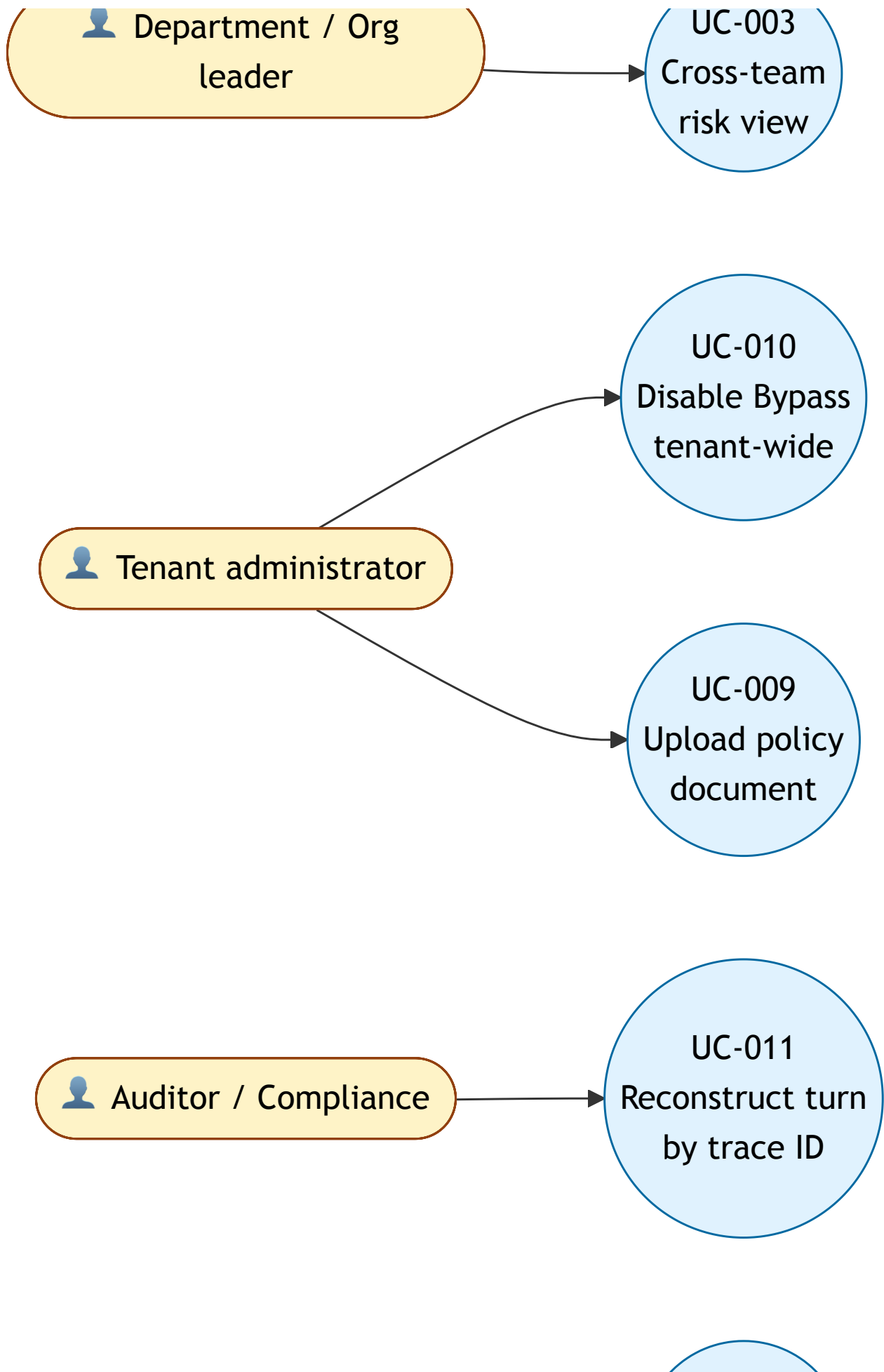
Each use case is referenced by an identifier (`UC-NNN`) and includes: actor, preconditions, primary scenario, alternative scenarios, postconditions, and the requirements it exercises. Use cases are illustrative; they do not exhaustively enumerate user behaviour but cover every primary capability area.

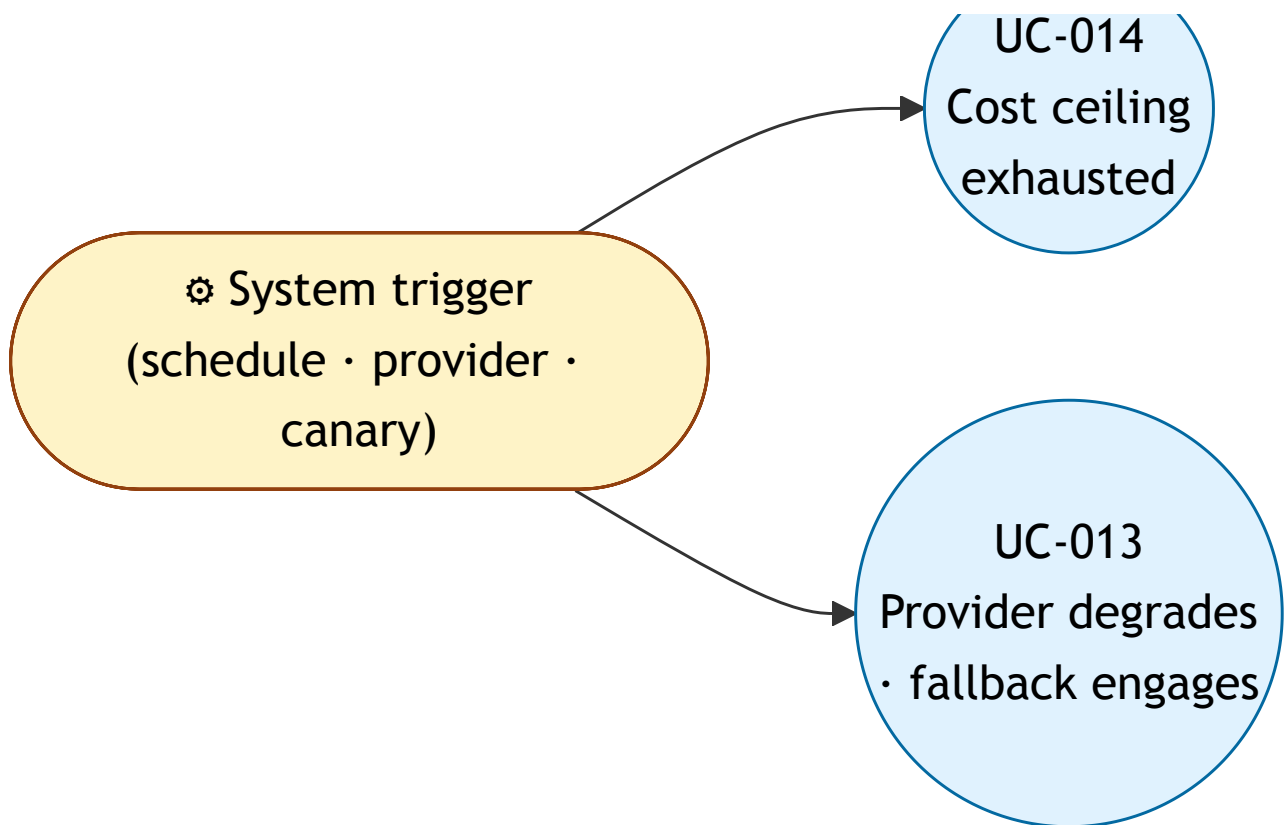
C.0 Use case diagram — Actors and use cases

The diagram below maps each external actor (§2.1) to the use cases they initiate. Actor-to-use-case lines indicate "this actor initiates this use case"; UML-style `«include»` and `«extend»` relationships are not shown — alternatives are captured in each use-case's narrative below.









How to read. Each circle is a use case detailed in §C.1 onward. Each rounded box is an actor from §2.1. A line from actor to use case means the actor is the primary initiator. Use cases without an actor line (none in Phase 1) would be system-internal flows.

UC-001 — Employee asks about open tasks

Field	Value
Actor	Employee
Preconditions	User is authenticated; conversation execution mode is set
Primary scenario	(1) User opens the global chat surface. (2) User asks "What are my open tasks this week?". (3) The agent classifies intent, retrieves tasks scoped to the user, composes a structured list with citations, and streams the response. (4) User clicks a citation to navigate to the source task.
Alternative — A	The user has no open tasks: agent responds with a plain-language "no open tasks" answer, no fabricated content (FR-038).
Alternative — B	LLM provider degrades during the response: ladder step engaged, user-visible notice shown (FR-036, NFR-029).
Postconditions	Audit event recorded; cost event accrued; conversation history updated.
Exercises	FR-001, FR-004, FR-005, FR-006, FR-019, FR-020, FR-028, FR-060, FR-061, NFR-001

UC-002 — Manager requests team status digest

Field	Value
Actor	Manager / Team lead
Preconditions	User has team-lead permission scope on at least one team
Primary scenario	(1) User asks "Give me a status digest for my team this sprint." (2) The agent identifies the user's overseen team, retrieves plan and task state, and composes a narrative-with-table response with on-track / at-risk / done breakdown and recent changes. (3) Citations link each claim to the source task or plan.
Alternative — A	The aggregated query falls below the minimum-group-size threshold for some metric: that metric is refused or k-anonymised (FR-025).
Postconditions	Audit event recorded; conversation updated.
Exercises	FR-001, FR-006, FR-025, FR-062, FR-063, FR-064

UC-003 — Department leader requests cross-team risk view

Field	Value
Actor	Department / Org leader
Preconditions	User has department-leader permission scope
Primary scenario	(1) User asks "Which cross-team dependencies look at risk in Q3?". (2) Agent retrieves plans across overseen teams, evaluates dependency status, composes a table with risk callouts and citations.
Postconditions	Audit event recorded.
Exercises	FR-019, FR-025, FR-063, FR-064

UC-004 — Employee creates a task in Default approvals mode

Field	Value
Actor	Employee
Preconditions	Conversation execution mode is Default approvals
Primary scenario	(1) User says "Create a task to follow up with Anh about the Q3 brief by Friday." (2) The agent resolves "Anh" via People, identifies parent plan from current screen / conversation, composes the structured intent, renders an inline preview card with Confirm / Decline. (3) User clicks Confirm. (4) The task is created; conversation reflects the outcome with a citation to the new task.
Alternative — A	User clicks Decline: the write is not executed; a structured decline event is recorded (FR-018).
Alternative — B	"Anh" is ambiguous (multiple matches): the agent surfaces the candidates and asks the user to choose before proceeding.
Postconditions	Audit event recorded with mode=default and confirmed_at timestamp; conversation updated.
Exercises	FR-008, FR-009, FR-018, FR-019, FR-020, FR-021, FR-065, FR-066, UI-007

UC-005 — Employee creates a task in Bypass approvals mode

Field	Value
Actor	Employee
Preconditions	Conversation execution mode is Bypass approvals; tenant has not disabled Bypass tenant-wide
Primary scenario	(1) User says "Reassign FUTURE-123 to Linh and push due date to next Tuesday." (2) The agent executes the mutation immediately under the user's permissions and includes the outcome in its reply with a citation to the updated task.
Alternative — A	Tenant has disabled Bypass tenant-wide: the request is treated as Default approvals (FR-015), inline preview shown.
Alternative — B	The tool is administrator-pinned to "always confirm" (FR-016): inline preview shown.
Postconditions	Audit event recorded with mode=bypass and no confirmed_at; conversation updated.
Exercises	FR-008, FR-010, FR-014, FR-015, FR-016, FR-067

UC-006 — Employee extracts meeting action items (always inbox)

Field	Value
Actor	Employee
Preconditions	User pastes or attaches a meeting transcript
Primary scenario	(1) User asks "Extract action items from this meeting transcript." (2) The agent extracts items, suggests an owner per item via People, attaches a confidence score, persists the batch as drafted tasks. (3) The platform notification module surfaces an approval-inbox card. (4) User opens the inbox, accepts or rejects per item. (5) Accepted items are executed under the user's authority.
Alternative — A	TTL elapses before user reviews the batch: drafts auto-reject; initiator is notified (FR-040, FR-041).
Alternative — B	Precondition revalidation fails for an accepted item (e.g. plan deleted): execution fails with a structured event; initiator notified (FR-042, FR-043).
Postconditions	Per-item audit events recorded; conversation updated with batch status.
Exercises	FR-012, FR-013, FR-040, FR-041, FR-042, FR-043, FR-068, FR-069, FR-070, UI-020, UI-021

UC-007 — Employee subscribes to a morning task brief

Field	Value
Actor	Employee
Preconditions	User is authenticated; tenant administrator has not capped the user's active schedules (FR-075)
Primary scenario	(1) User opens the schedules surface. (2) User configures a morning task brief at 08:00 weekdays. (3) The platform creates a delegation grant scoped to the brief's read intents. (4) Each weekday morning the schedule fires; the agent runs read-only, produces the brief, and the notification module delivers it.
Alternative — A	Delegation grant has expired: the run is refused; user is notified (FR-072).
Alternative — B	Per-delegation budget exhausted: the run is refused with a budget reason (FR-077, NFR-016).
Postconditions	Schedule run audit recorded.
Exercises	FR-003, FR-026, FR-027, FR-071, FR-072, FR-073, FR-074

UC-008 — Employee asks an HR-handbook question (knowledge base)

Field	Value
Actor	Employee
Preconditions	The tenant has uploaded the handbook to its knowledge base
Primary scenario	(1) User asks "How many days of bereavement leave am I entitled to?". (2) The agent retrieves passages from the tenant knowledge base. (3) The agent composes an answer with citations linking to the source document and section.
Alternative — A	Embedding API degraded: retrieval returns empty; agent answers without the knowledge base and discloses the gap (FR-039).
Alternative — B	No relevant passage found: agent responds with "I could not find this in your knowledge base" plainly (FR-038).
Postconditions	Audit event recorded.
Exercises	FR-006, FR-038, FR-039, FR-050, FR-051, FR-052

UC-009 — Tenant administrator uploads a policy document

Field	Value
Actor	Tenant administrator
Preconditions	Administrator role granted; tenant under document quota (FR-057); document under per-document size cap (FR-058)
Primary scenario	(1) Administrator opens the knowledge-base management view. (2) Administrator uploads a PDF policy document. (3) The system asynchronously chunks, embeds, indexes the document. (4) Notification delivered to administrator on completion. (5) Document is browsable and retrievable.
Alternative — A	Document exceeds size cap: upload rejected at upload time with a structured reason.
Alternative — B	Ingestion fails (e.g. image-PDF, OCR not supported at Phase 1, CN-10): document marked failed with a structured failure reason; administrator notified (FR-056, FR-059).
Alternative — C	Document successfully ingested but later deprecated by administrator: deprecated document excluded from retrieval (FR-055).
Postconditions	Ingestion run audit recorded; document available for retrieval.
Exercises	FR-053, FR-054, FR-055, FR-056, FR-057, FR-058, FR-059, NFR-006, UI-017

UC-010 — Tenant administrator disables Bypass mode tenant-wide

Field	Value
Actor	Tenant administrator
Preconditions	Administrator role granted
Primary scenario	(1) Administrator opens execution-mode policy view. (2) Administrator toggles "Disable Bypass tenant-wide" and saves. (3) Within five minutes, all conversations in the tenant behave as Default mode regardless of user selection.
Postconditions	Configuration change audit recorded with old value, new value, administrator identity, timestamp.
Exercises	FR-015, FR-080, FR-082, FR-083, NFR-038

UC-011 — Auditor reconstructs a turn by trace identifier

Field	Value
Actor	Auditor / Compliance reviewer
Preconditions	Auditor role granted; trace identifier available (e.g. from an incident report)
Primary scenario	(1) Auditor enters trace identifier into the audit query interface. (2) The system returns: initiator identity, delegation reference, execution mode, sequence of tool calls, prompt-store hashes, output reference, timestamps, and any confirmation events. (3) Auditor invokes "replay" to reconstruct the prompt and tool-call sequence deterministically.
Alternative — A	Prompt-store hash cannot be resolved: the system raises an error and refuses to fall back to a fuzzy match (FR-049).
Postconditions	Auditor obtains the full lineage in a single query (FR-022); reconstruction is identical to the original turn (FR-046).
Exercises	FR-020, FR-021, FR-022, FR-046, FR-047, FR-048, FR-049

UC-012 — Employee cancels an in-flight turn

Field	Value
Actor	Employee
Preconditions	A turn is in flight
Primary scenario	(1) User clicks Cancel. (2) The cancellation signal propagates within the sub-second target. (3) The agent ceases at the next ceasing point; conversation is updated with a cancellation indicator and reason <code>user</code> .
Alternative — A	A write committed before the cancellation reached the gateway: the agent presents the truthful timestamp of the commit (FR-024); no fictional rollback message.
Postconditions	Cancellation audit recorded with reason and timestamp.
Exercises	FR-023, FR-024, FR-085, FR-086, FR-087, NFR-008, NFR-035

UC-013 — LLM provider degrades; fallback engages

Field	Value
Actor	Any user
Preconditions	LLM provider is in a degraded state (transient errors or quality-canary flagged)
Primary scenario	(1) User asks a question. (2) Provider call returns transient error; runtime retries once, then falls back to the smaller model. (3) The agent emits a user-visible notice that the experience has changed and continues to answer. (4) Quality canary detects continued degradation and routes the tenant to the smaller tier.
Alternative — A	Both tiers degrade severely (success rate < 50%): the system refuses new turns with the elevated-degradation message (NFR-029, FR-037).
Alternative — B	Tenant daily budget reached during degradation: refuse with budget reason (FR-077, NFR-029).
Postconditions	Degradation event audited; trace tagged with the ladder step reached.
Exercises	FR-036, FR-037, FR-038, NFR-027, NFR-028, NFR-029, NFR-030

UC-014 — Cost ceiling exhausted; refusal with reason

Field	Value
Actor	Employee or asynchronous run
Preconditions	Tenant or per-user daily budget at 100%
Primary scenario	(1) User submits a turn (or a schedule fires). (2) The pre-turn budget check refuses; user / schedule owner sees a plain-language refusal with the budget reason. (3) Audit event recorded.
Alternative — A	Budget reaches 80% (warning threshold): the user sees a non-blocking notice at turn start; turns continue.
Postconditions	Refusal audit recorded; cost-event ledger reflects the day's consumption.
Exercises	FR-037, FR-077, NFR-004, NFR-016

Appendix D. Requirements Traceability Matrix

The matrix below summarises requirement coverage by feature area. Every requirement in §3 and §4 is uniquely identified; tooling can mechanically extract this matrix from this document.

Feature / Area	Requirement IDs	Use Cases
Conversational surfaces and output	FR-001 – FR-007	UC-001, UC-002, UC-008
Execution modes and approval routing	FR-008 – FR-018	UC-004, UC-005, UC-006
Governance and trust boundary	FR-019 – FR-027	UC-001, UC-003, UC-004, UC-011
Conversation memory	FR-028 – FR-035	UC-001, UC-008
Honesty and degradation	FR-036 – FR-039	UC-008, UC-013
Approval lifecycle	FR-040 – FR-045	UC-006
Audit trail and deterministic replay	FR-046 – FR-049	UC-011
Tenant knowledge base	FR-050 – FR-059	UC-008, UC-009
Planner — read capabilities	FR-060 – FR-064	UC-001, UC-002, UC-003
Planner — write capabilities	FR-065 – FR-070	UC-004, UC-005, UC-006
Scheduled and event-triggered runs	FR-071 – FR-075	UC-007
Tenant administration	FR-076 – FR-084	UC-009, UC-010
Cancellation and abort	FR-085 – FR-088	UC-012, UC-013
User interfaces	UI-001 – UI-025	UC-001, UC-004, UC-006, UC-009, UC-010
External interfaces	EI-001 – EI-016	All
Performance	NFR-001 – NFR-008	UC-001, UC-006
Security and privacy	NFR-009 – NFR-019	UC-001, UC-003, UC-011
Usability	NFR-020 – NFR-025	All
Reliability	NFR-026 – NFR-035	UC-006, UC-012, UC-013
Maintainability and scalability	NFR-036 – NFR-039	UC-010
Database	DB-001 – DB-011	UC-009, UC-011

Legal, regulatory, compliance	LR-001 – LR-014	UC-009, UC-011
Internationalisation and localisation	I18N-01 – I18N-07	All

Document Control

Field	Value
Version	1.0 (baseline)
Status	Approved baseline
Date	2026-05
Owner	Future platform engineering — Agents track
Distribution	Internal — engineering, product, compliance, operations, executive sponsors
Change control	Any change to a numbered requirement is a documented amendment with a new revision and approval.
Next review	At Phase 1 acceptance; thereafter quarterly while the Agents module is active.